

Lattice QCD for precision and fundamental symmetry tests of the Standard Model

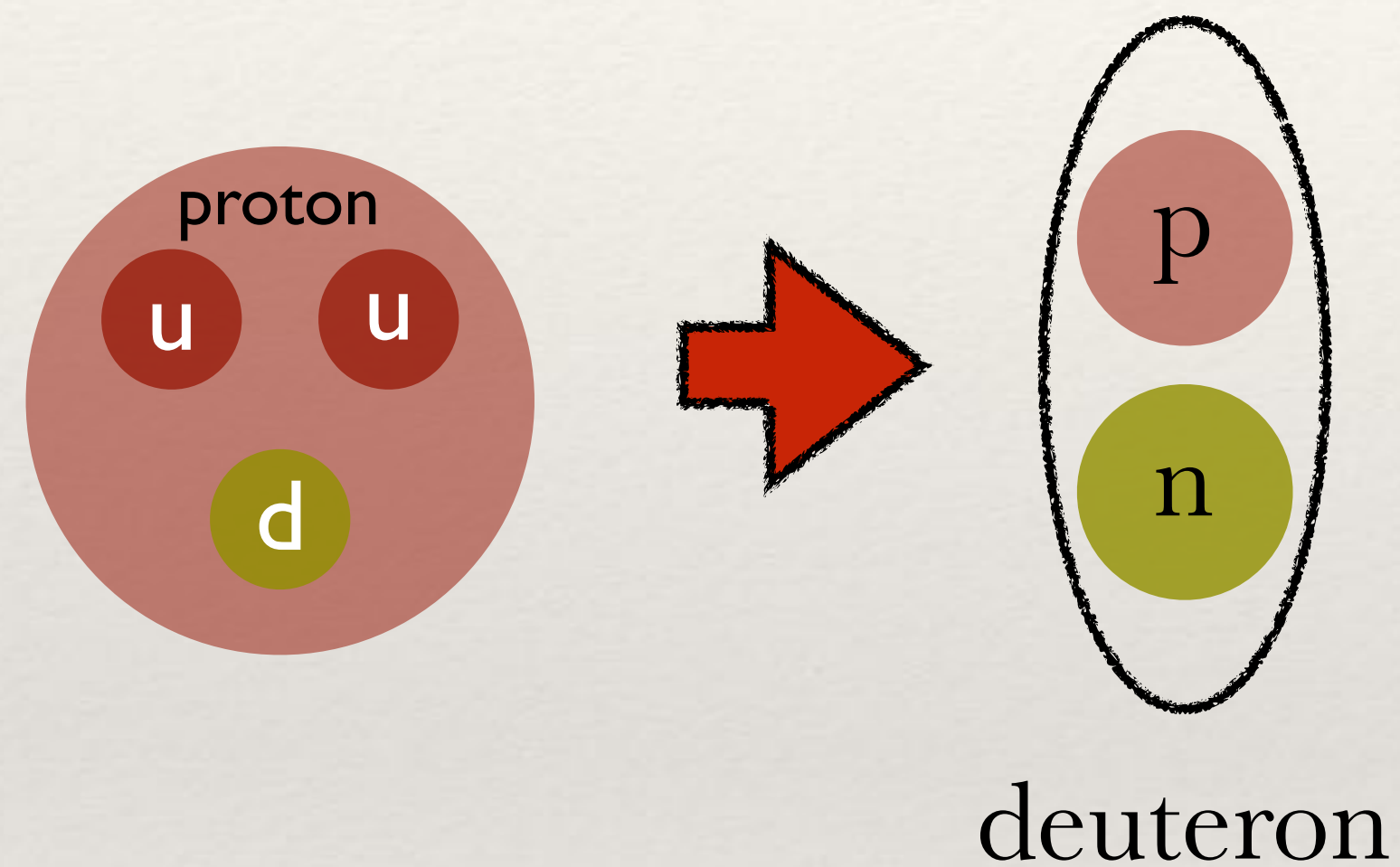
SciDAC 2026 PI Meeting
4-6 August

André Walker-Loud



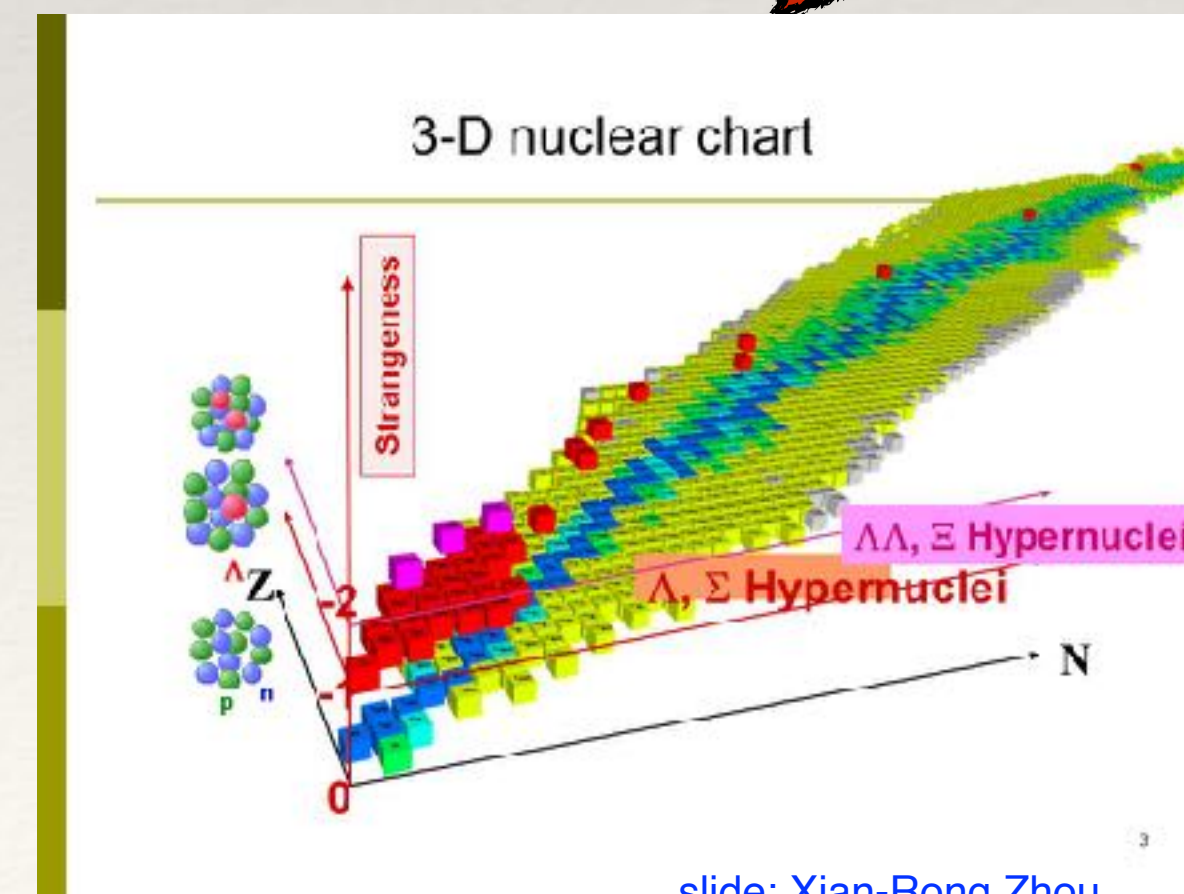
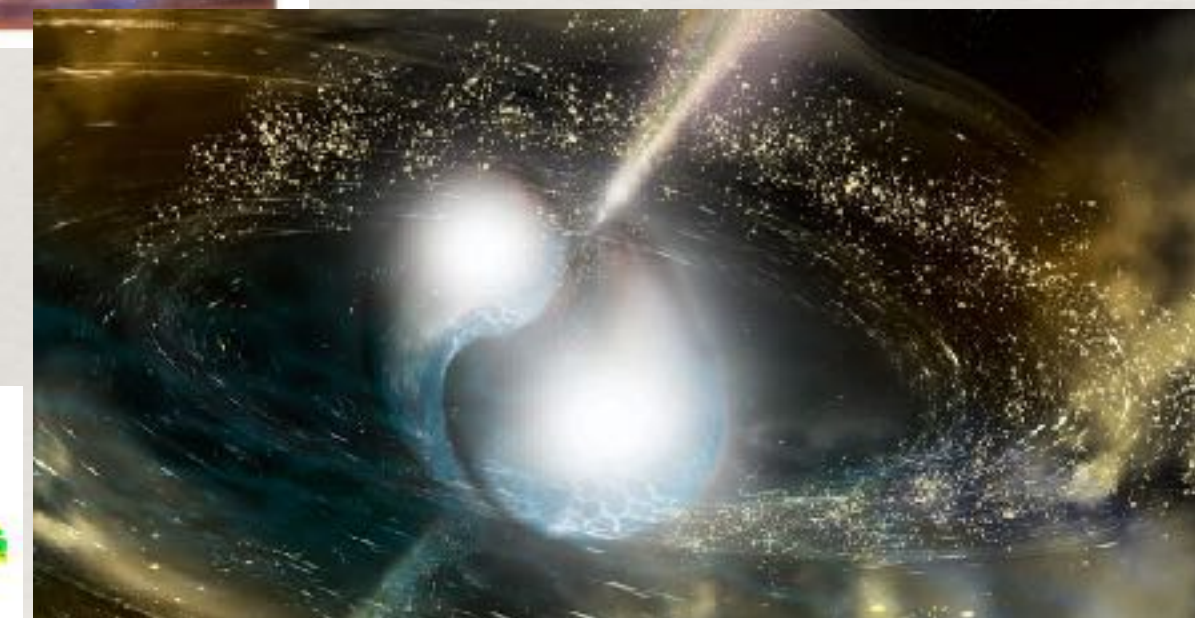
What are the science drivers?

❑ Can we **quantitatively understand** nuclear physics from the fundamental degrees of freedom?



Does **strange matter** play an important role in neutron stars?

opportunity to work with **NUCLIE+ENAF**



What are the science drivers?

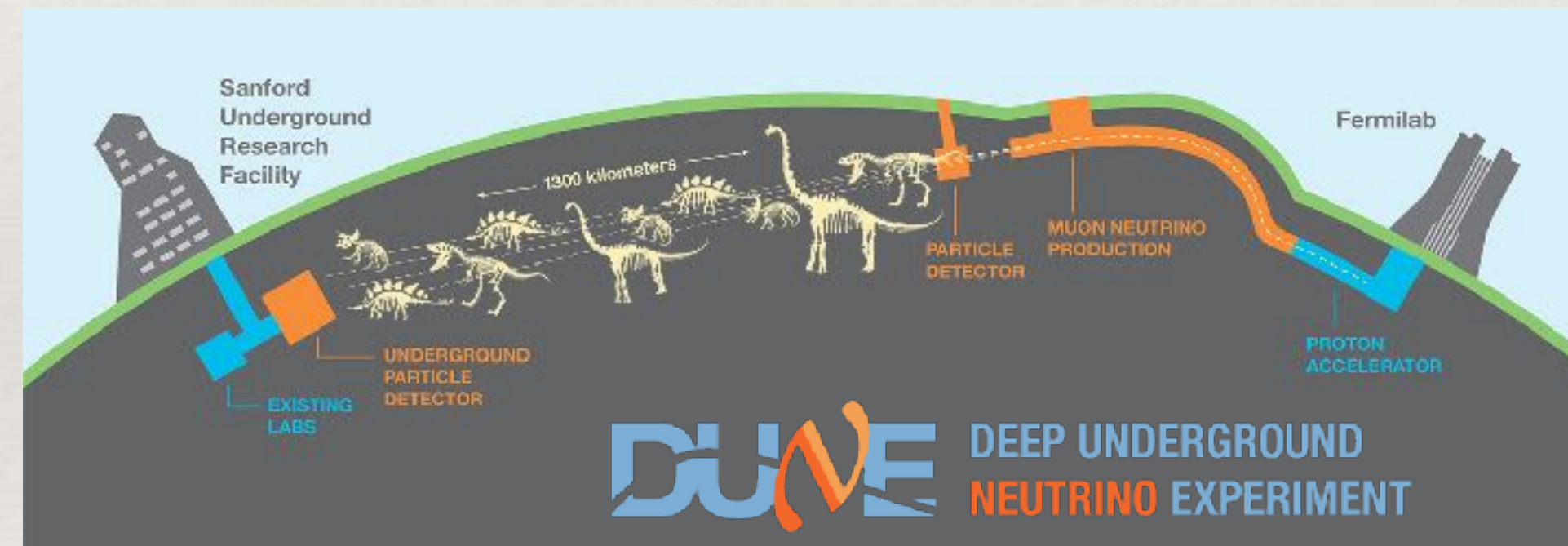
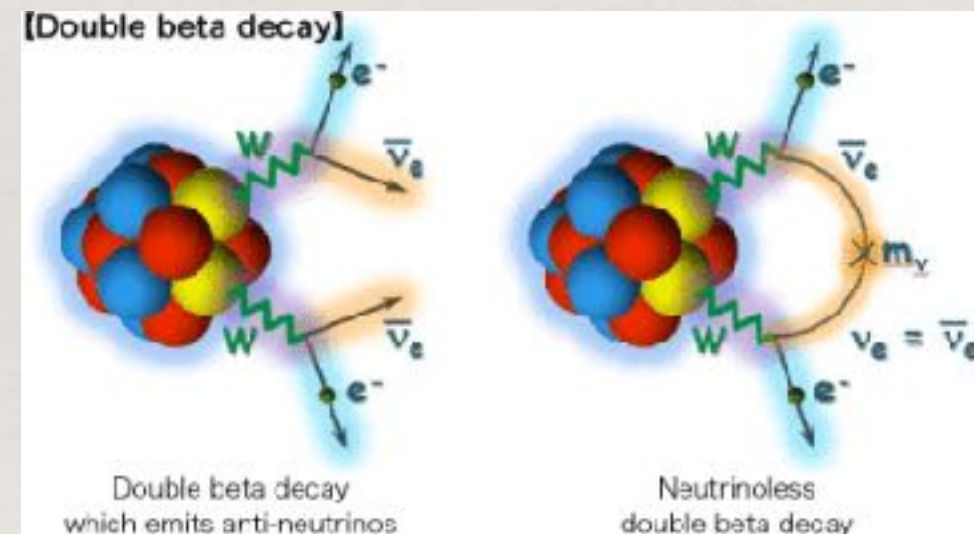
□ Precision and fundamental symmetry tests of the Standard Model of physics

□ Why is the universe made of matter

(and not anti-matter)?

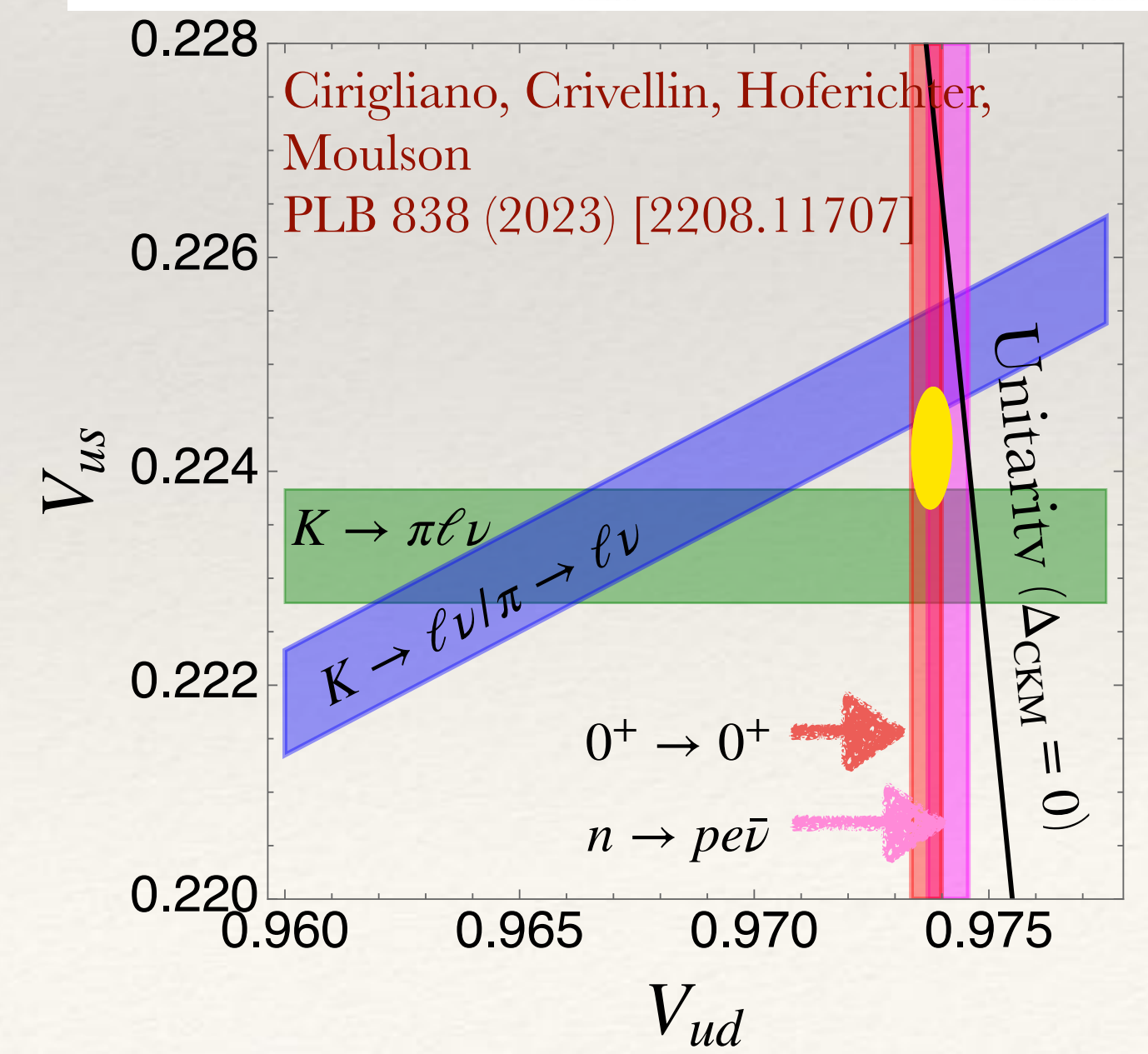
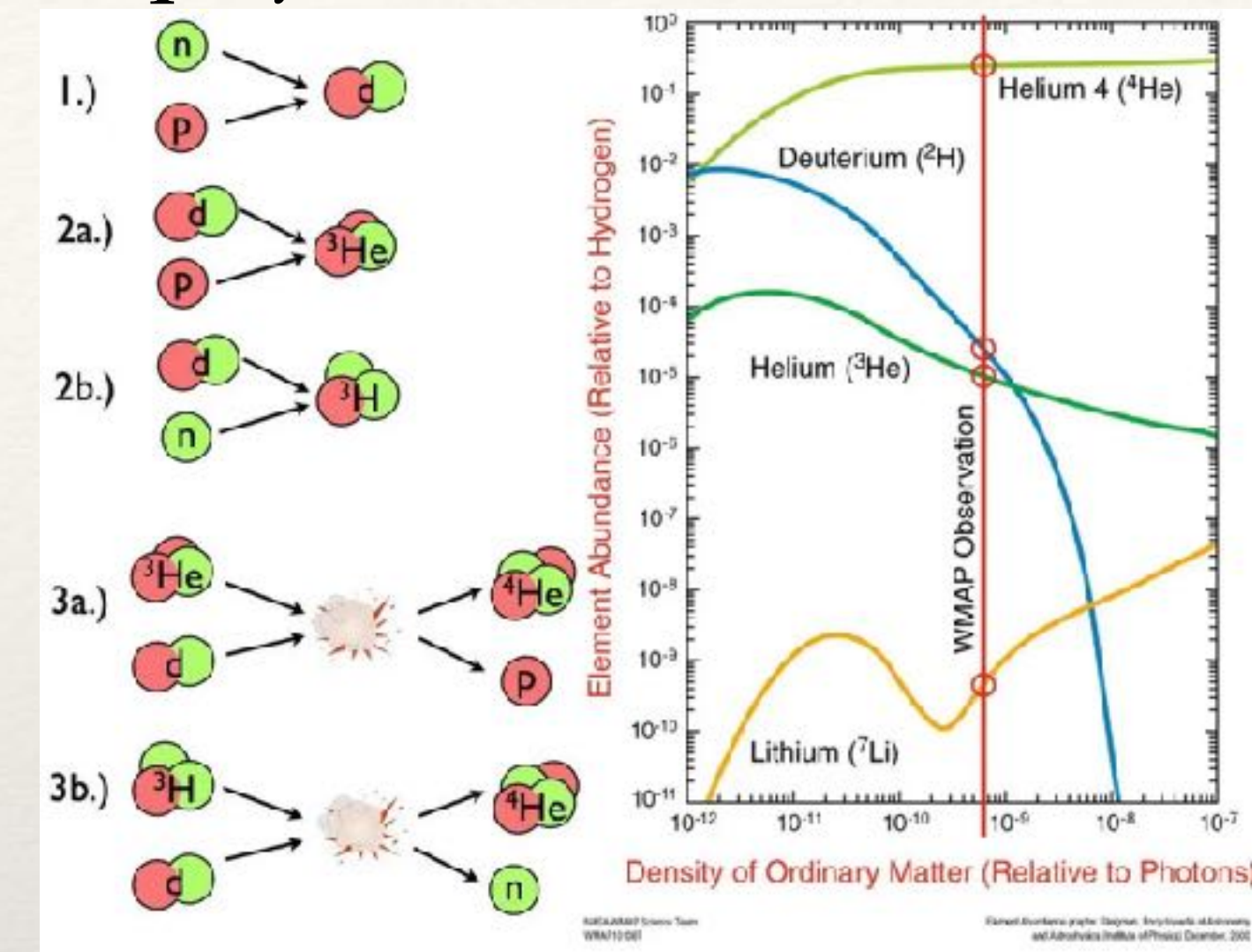
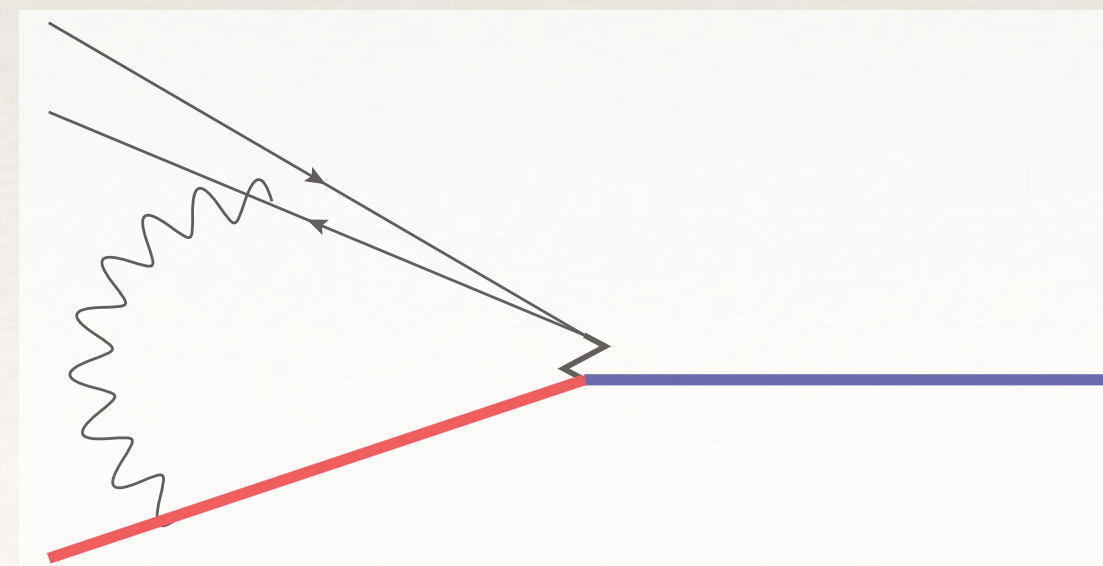
$$\frac{N_{p,n}}{N_\gamma} \approx 6 \times 10^{-10}$$

□ What is the nature of the neutrino and why is its mass so small?

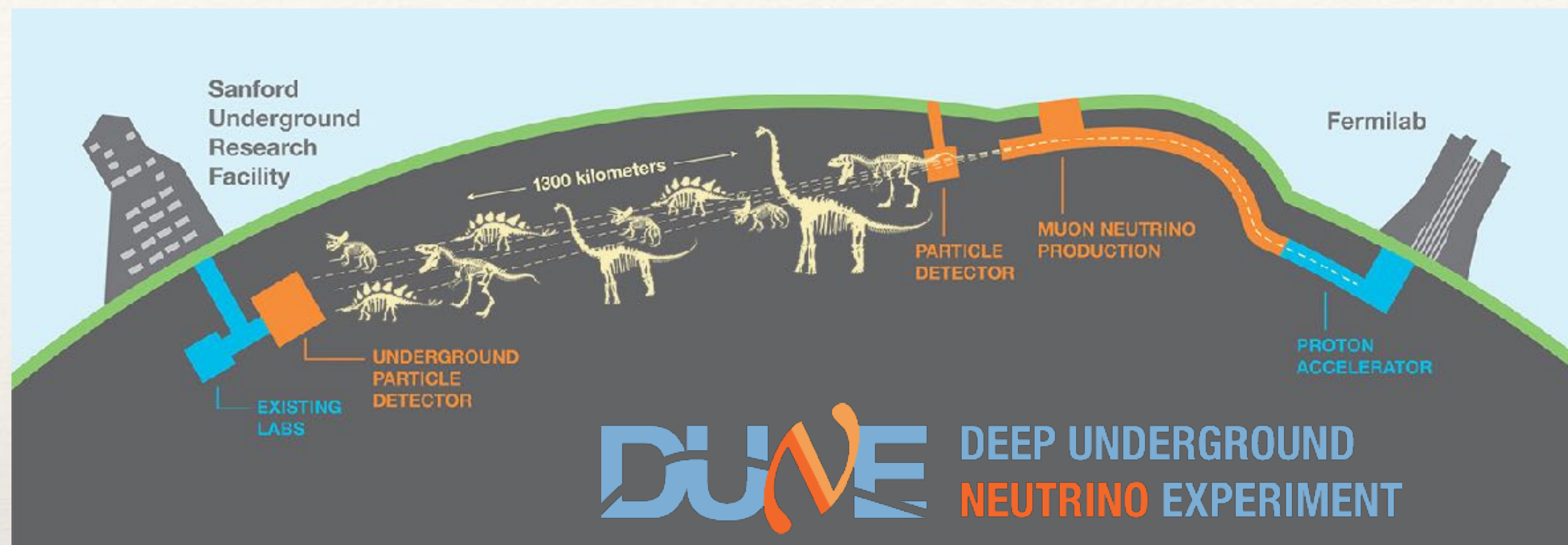


□ Can we detect signatures of new physics in precise, low-energy tests?

β -decay involves all 3 forces of Standard Model



$\nu - A$ Interactions



- ❑ Imperfect analogy
- ❑ If we aim to find new physics, it is wise to have accurate predictions of the “background” of known physics
- ❑ “evidence” of supersymmetry was in fact QCD jet physics

Event rate

Neutrino flux

Cross section

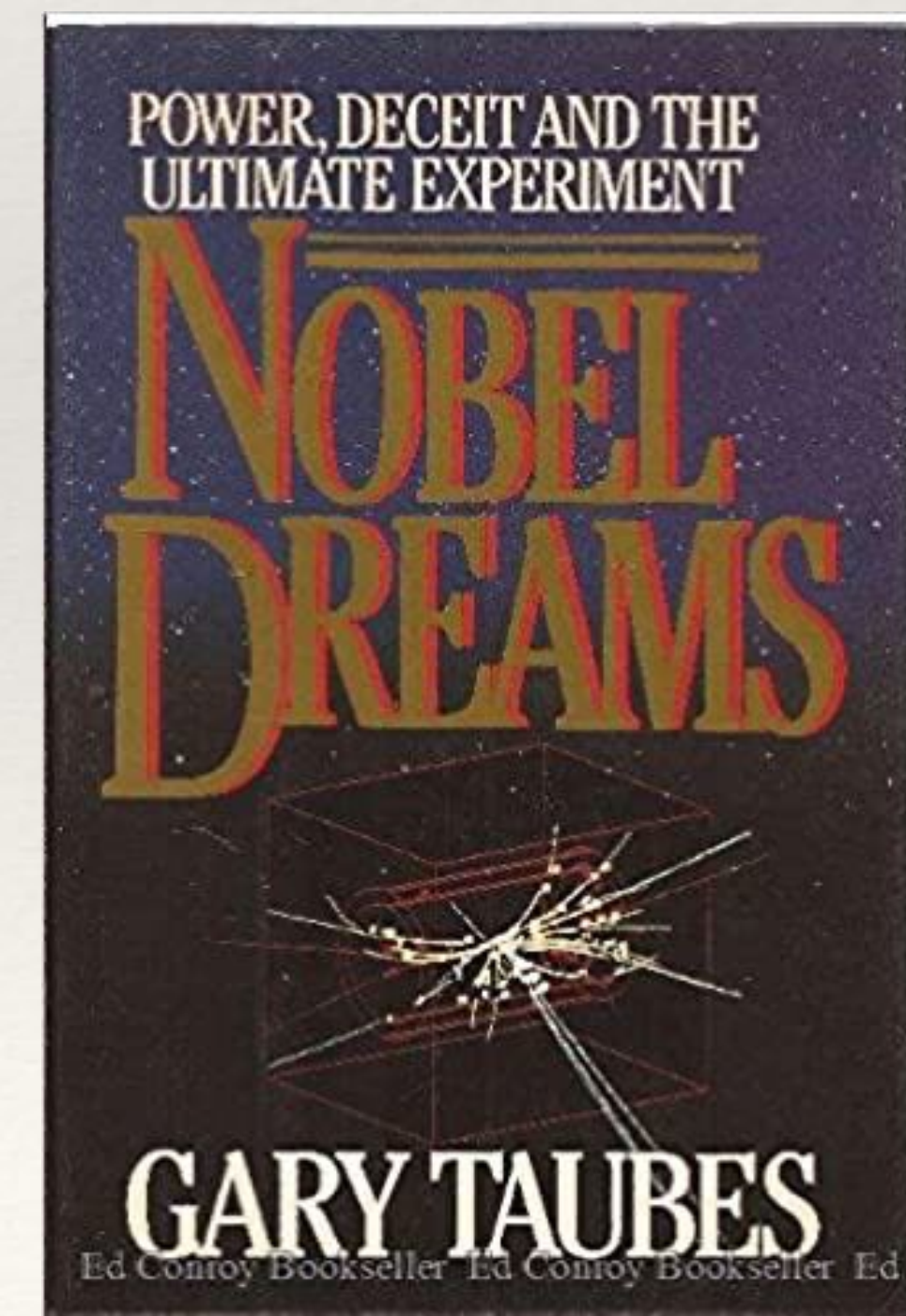
Detector smearing

Oscillation probability

$$R(\vec{x}) = \int dE \underbrace{\Phi(E_\nu) \times \sigma(E_\nu, \vec{x}) \times \epsilon(\vec{x}) \times P(E_\nu; \nu_A \rightarrow \nu_B)}_{\text{Far}}$$

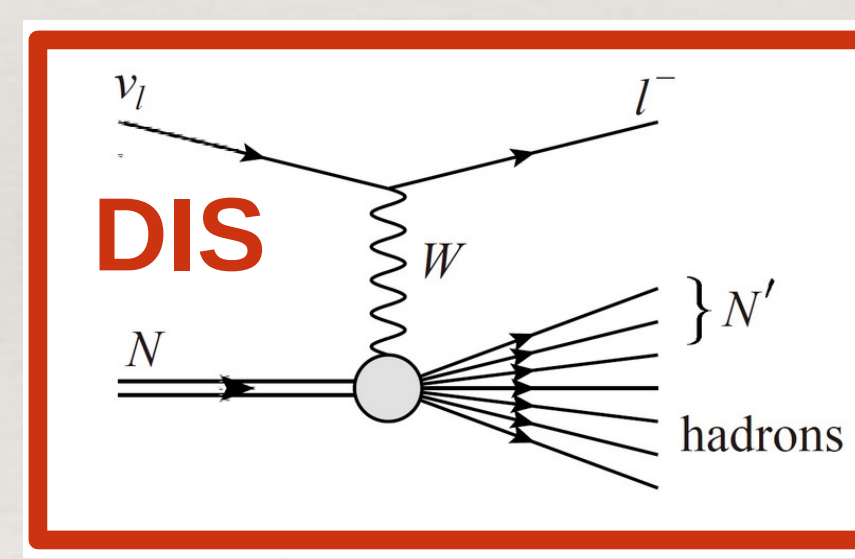
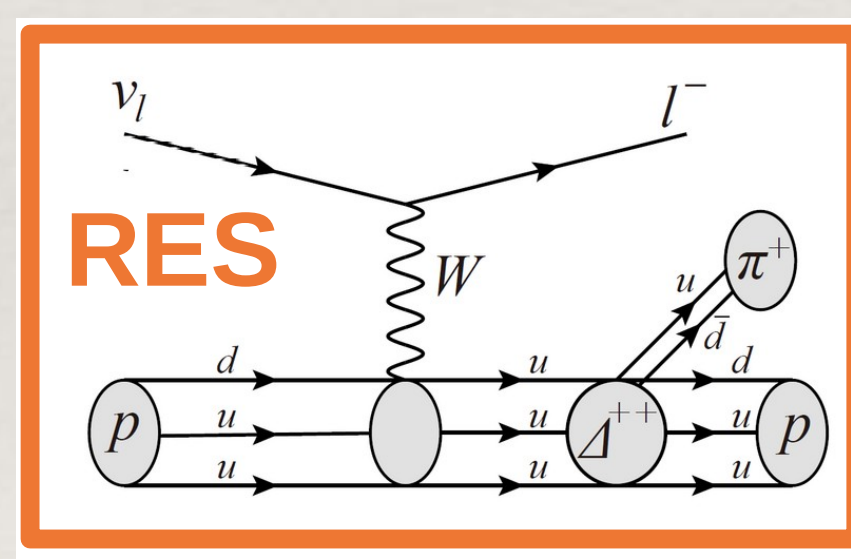
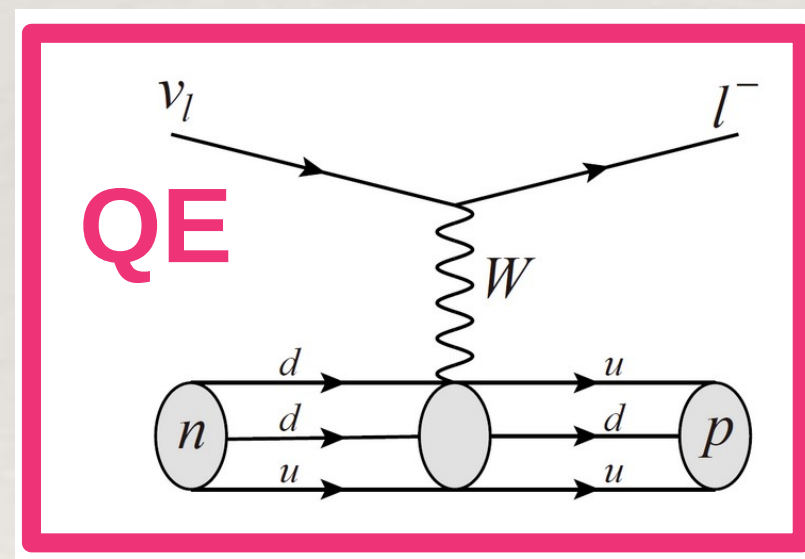
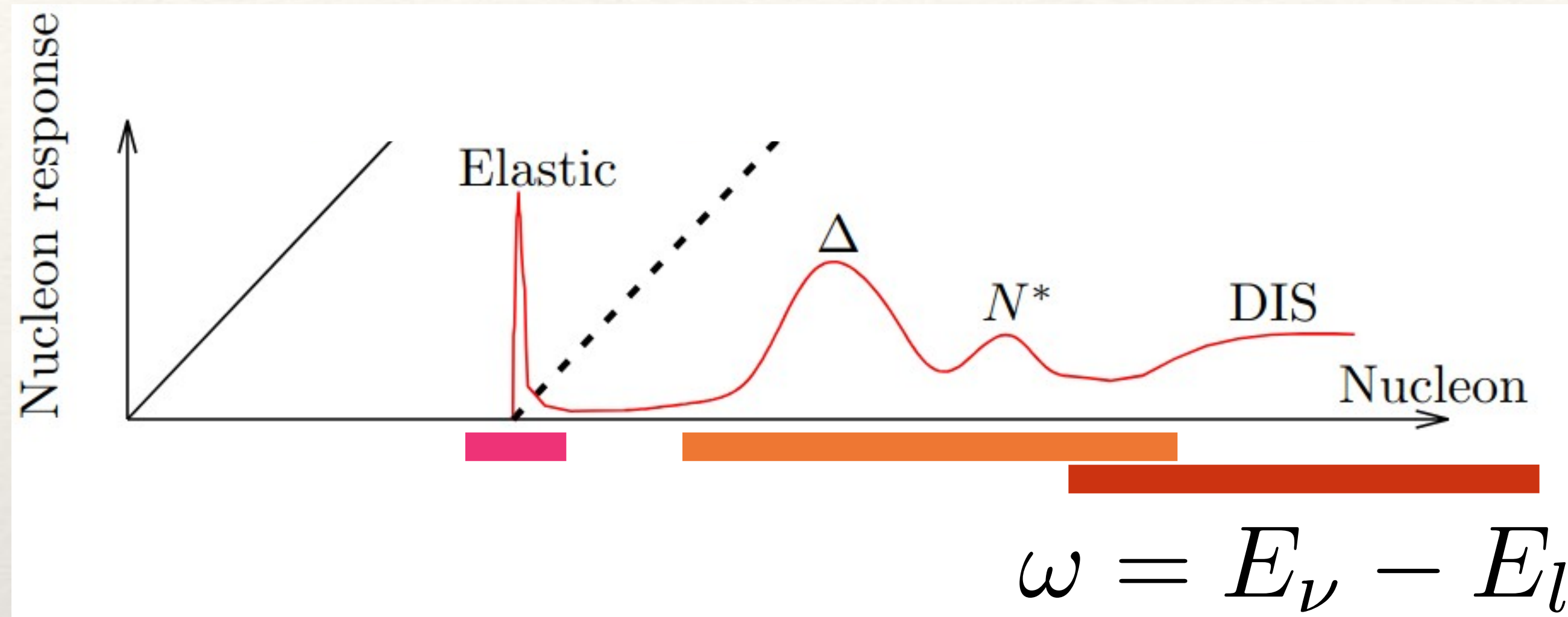
Near

- ❑ Challenges with all these pieces, but as theorists, we’ll focus on the cross-section



$\nu - A$ Interactions

□ Single nucleon response



□ Quasi Elastic (QE) Region

“On-shell” initial neutron and final proton (nuclear effects keep the nucleons slightly off-shell)

□ Resonance (RES) Region

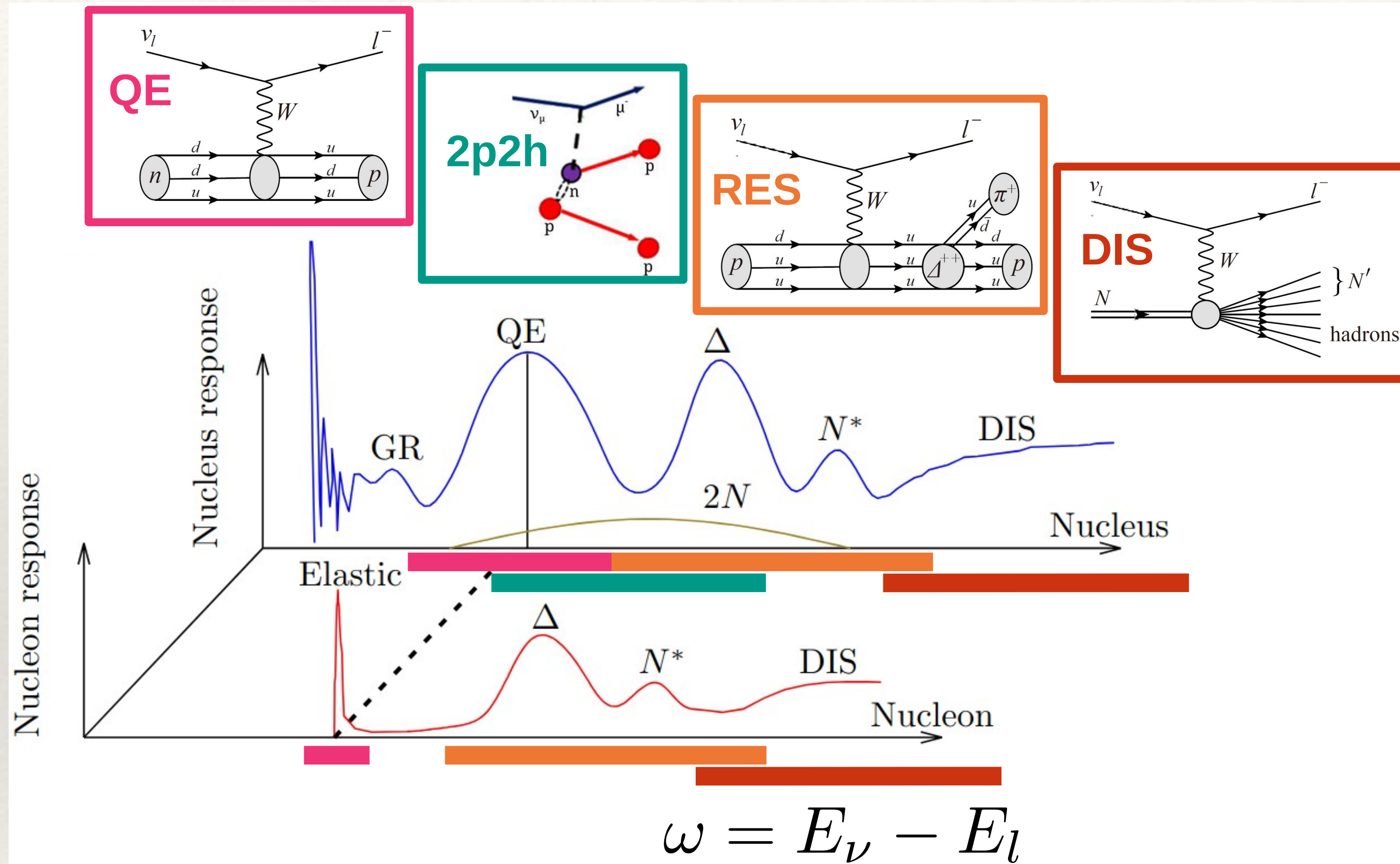
struck nucleon resonates inelastically decaying by emitting a pion, photon (multiple pions) which can strike other nucleons or escape the nucleus

□ Deep Inelastic Scattering (DIS) Region

The neutrino strikes individual partons in the nucleon

$\nu - A$ Interactions

□ “Nuclear” Response



□ Nuclear Excitations

□ Quasi Elastic (QE) Region

The nuclear QE response is smeared out from Fermi motion of the nucleons

□ 2-particle 2-hole (2p2h)

Leading many-body nuclear corrections (2-body currents)

□ Resonance (RES) Region

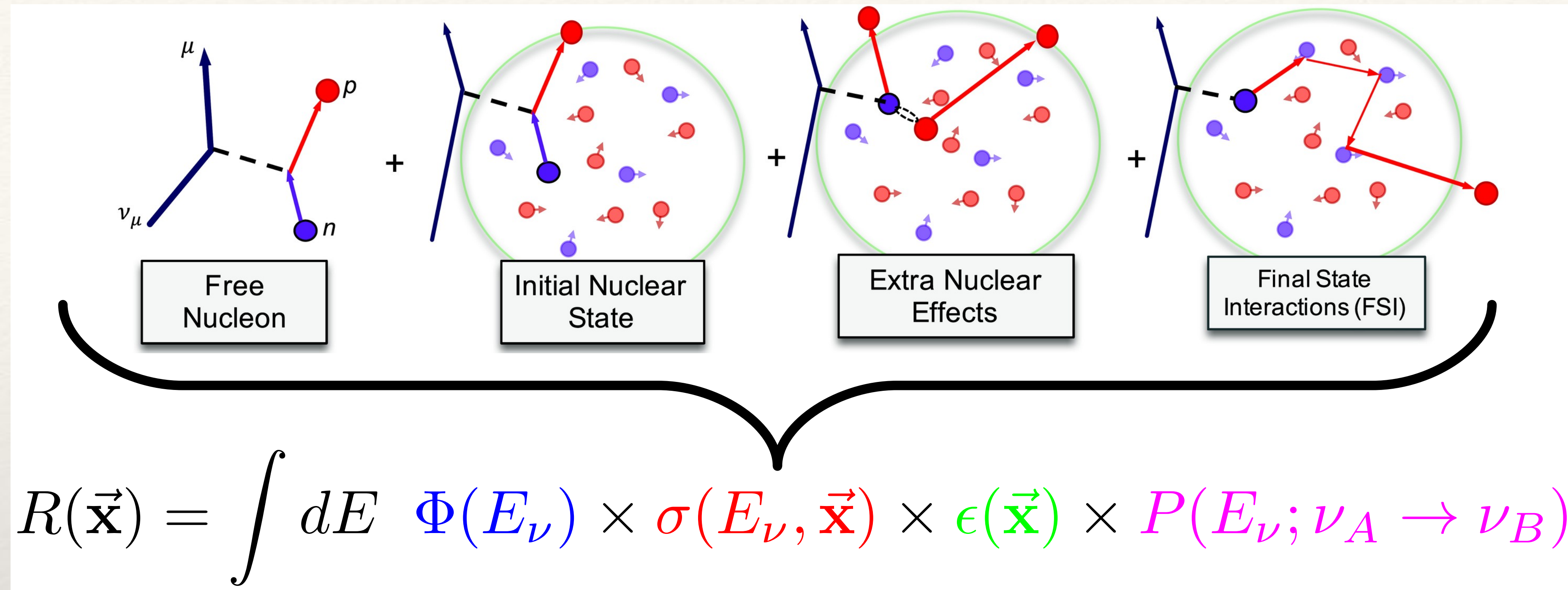
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□ Deep Inelastic Scattering (DIS) Region

The neutrino strikes individual partons in the nucleon

$\nu - A$ Interactions

❑ It's worse than that...



❑ Final State Interactions (FSI) really “gum up” the modeling of the nuclear response

❑ Outgoing particle content does NOT isolate what particular interaction has happened in nucleus

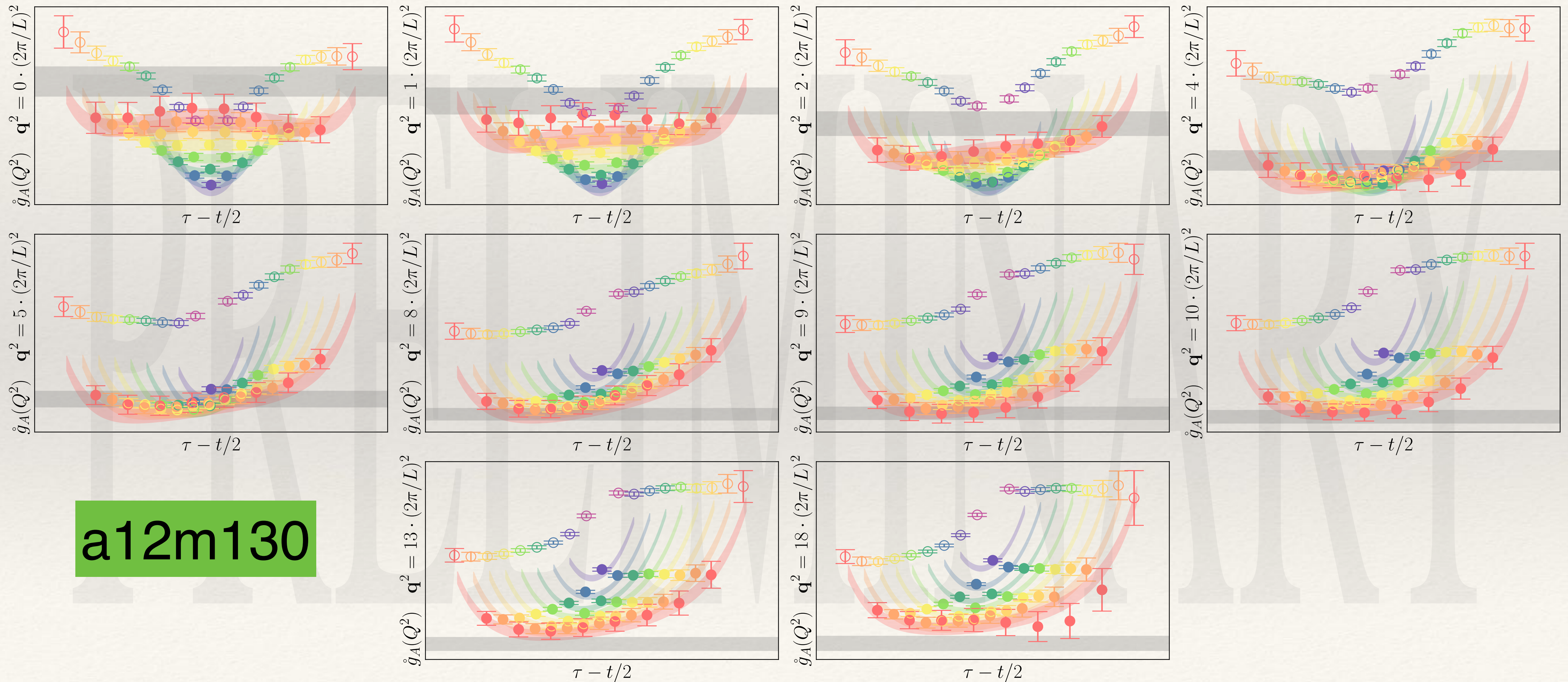
❑ Modeling uncertainty of the nuclear cross section injects uncertainty into the extraction of P_{osc} parameters

❑ Can it be controlled? We want to provide robust predictions of pieces of interaction — reducing space of parameters that require modeling, thus reducing uncertainty

LQCD: Single Nucleon Axial Form Factor

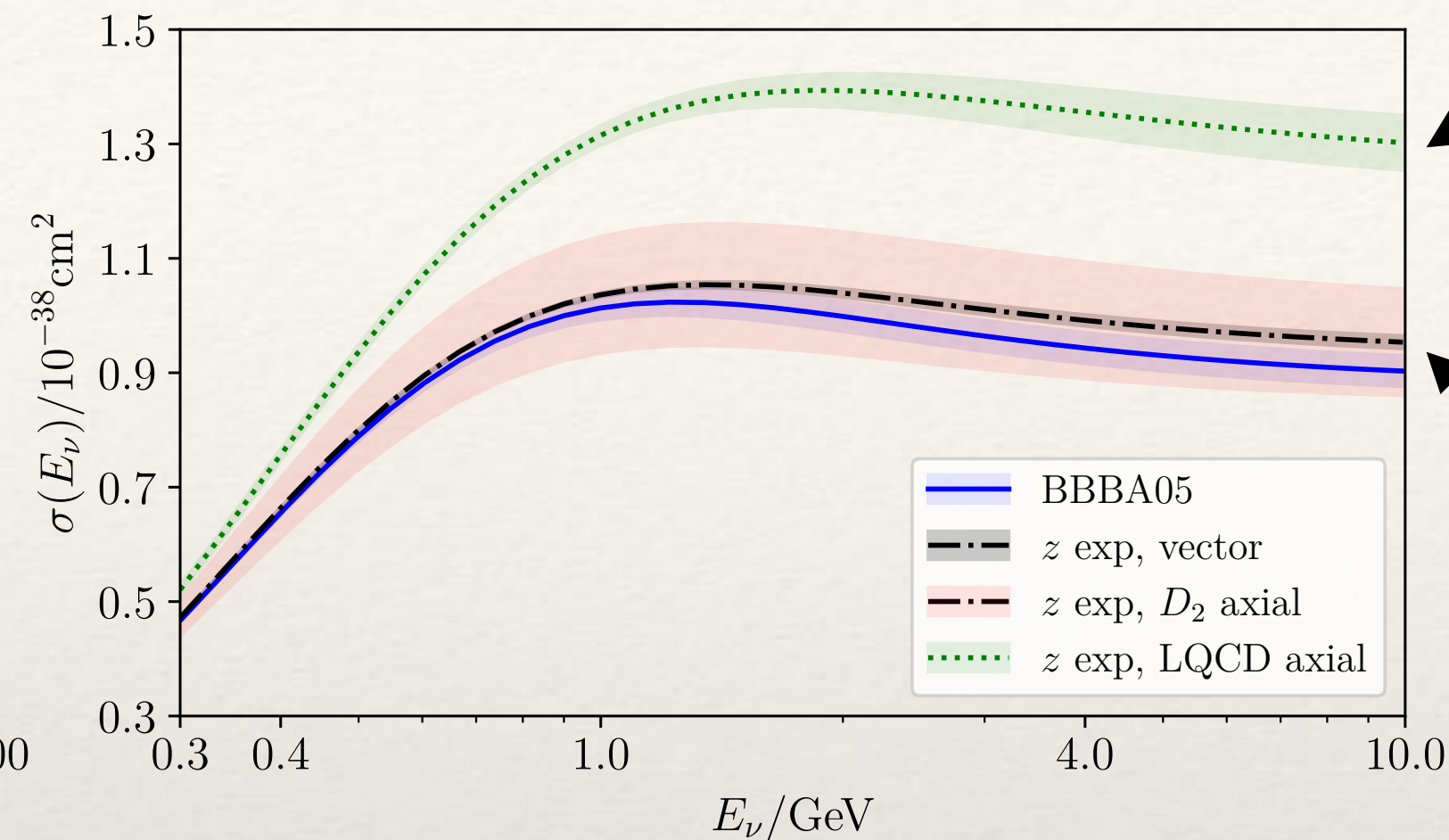
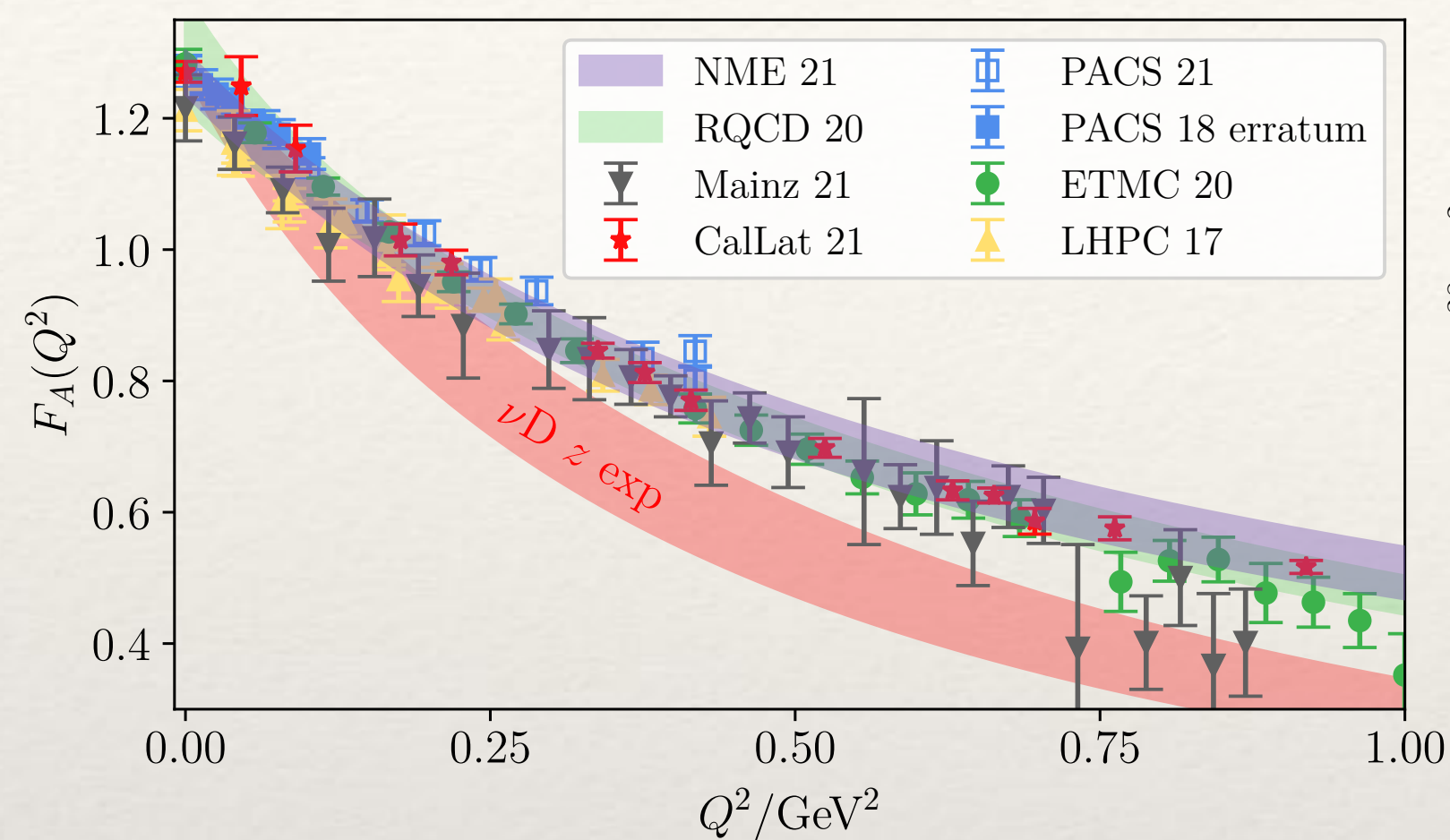
- zero-momentum sink ($p=0$), non-zero momentum transfer ($q \neq 0$)
source has momentum $-q$ by momentum conservation

[2111.06333]



LQCD: Single Nucleon Axial Form Factor

Meyer, Walker-Loud, Wilkinson
Ann. Rev. Nucl. Part. Sci. 72 (2022)



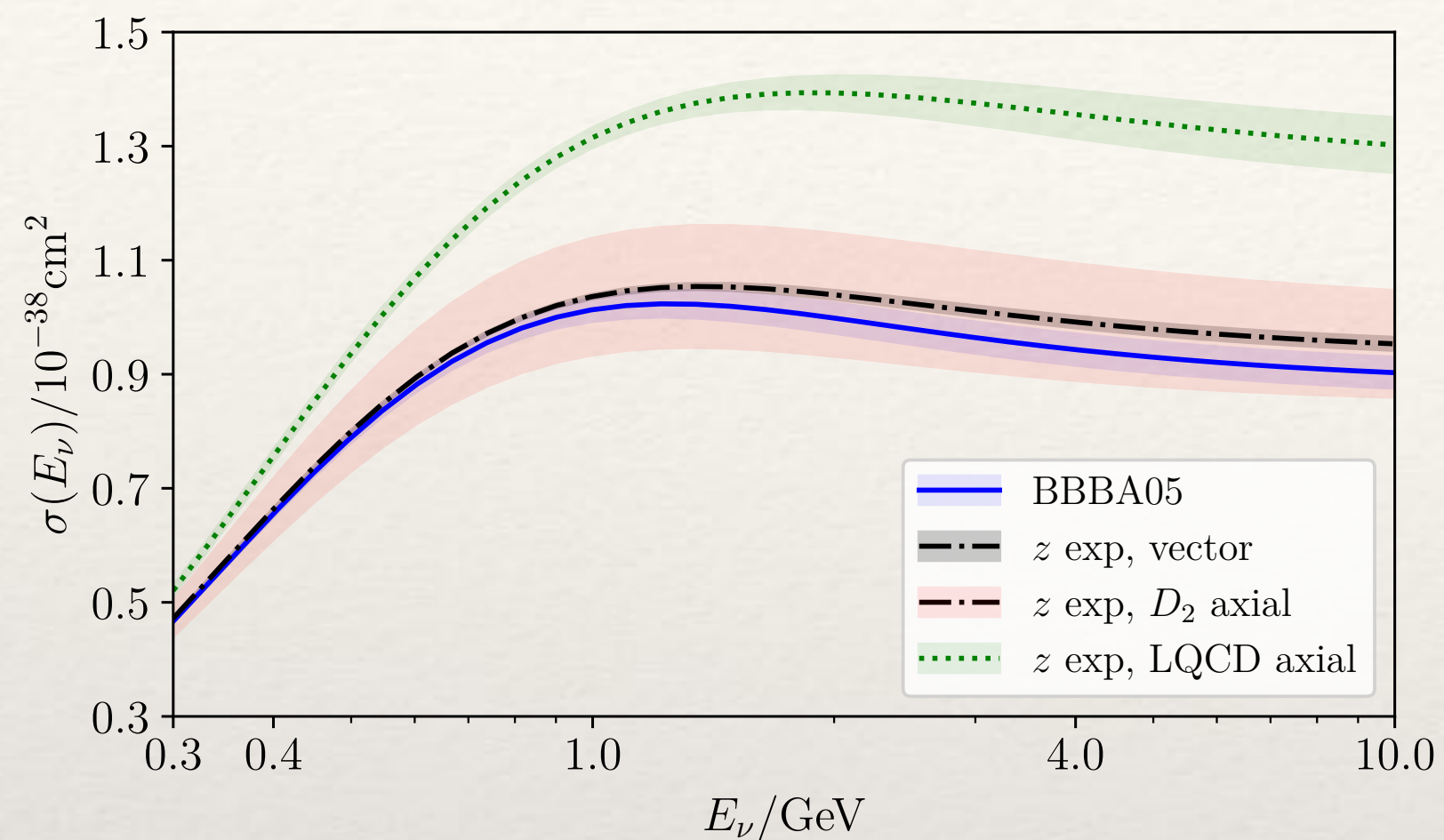
This discrepancy impacts the ability to interpret neutrino oscillation parameters

Some of this uncertainty comes from nucleon electromagnetic form factors

- ❑ Lattice QCD determination of $F_A(Q^2)$ is **inconsistent** with older phenomenological extraction
- ❑ results in 30% increase in ν -N cross section
- ❑ New MINER ν A measurement of νN with hydrocarbon lies between LQCD and νD
Nature 614 (2023)

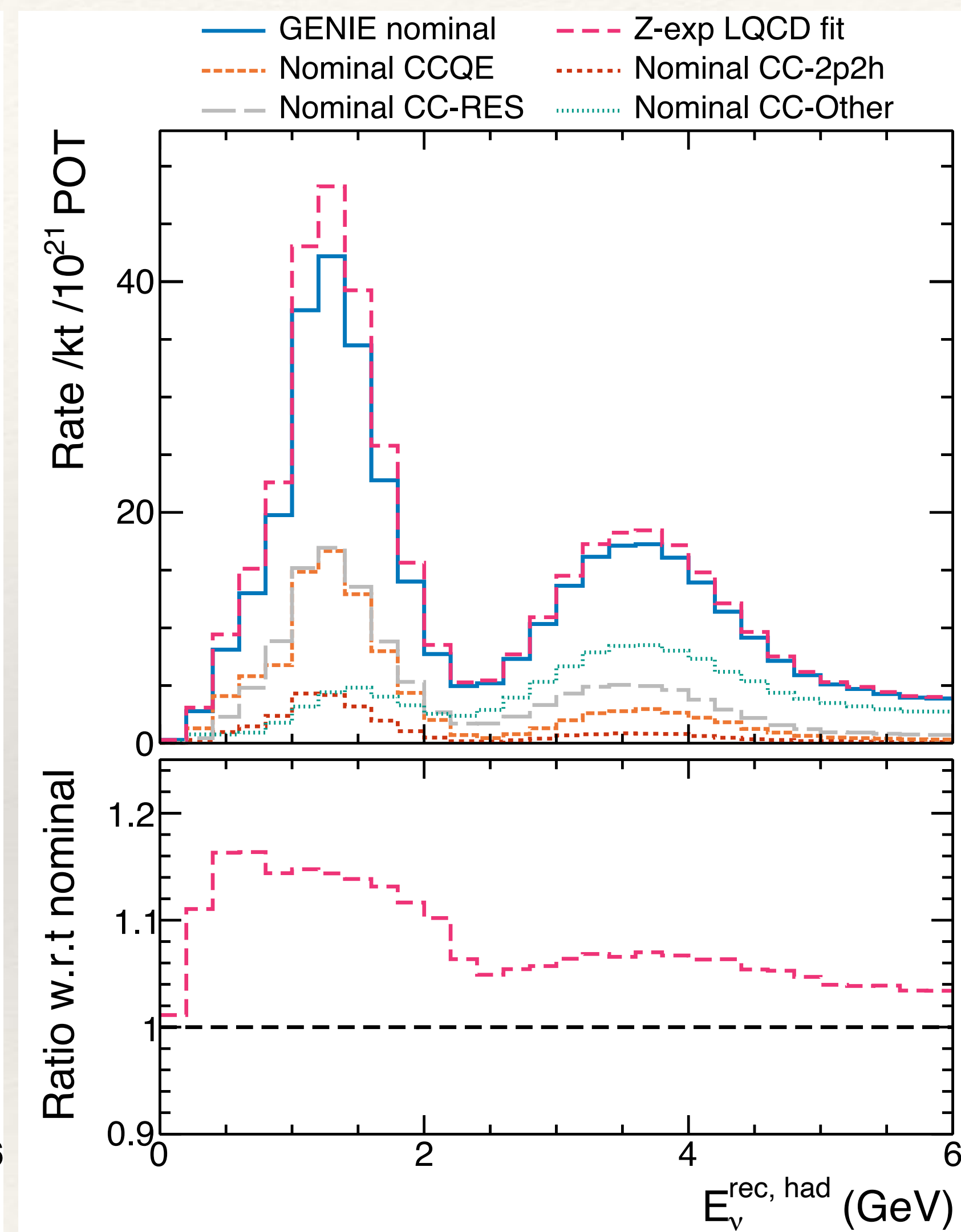
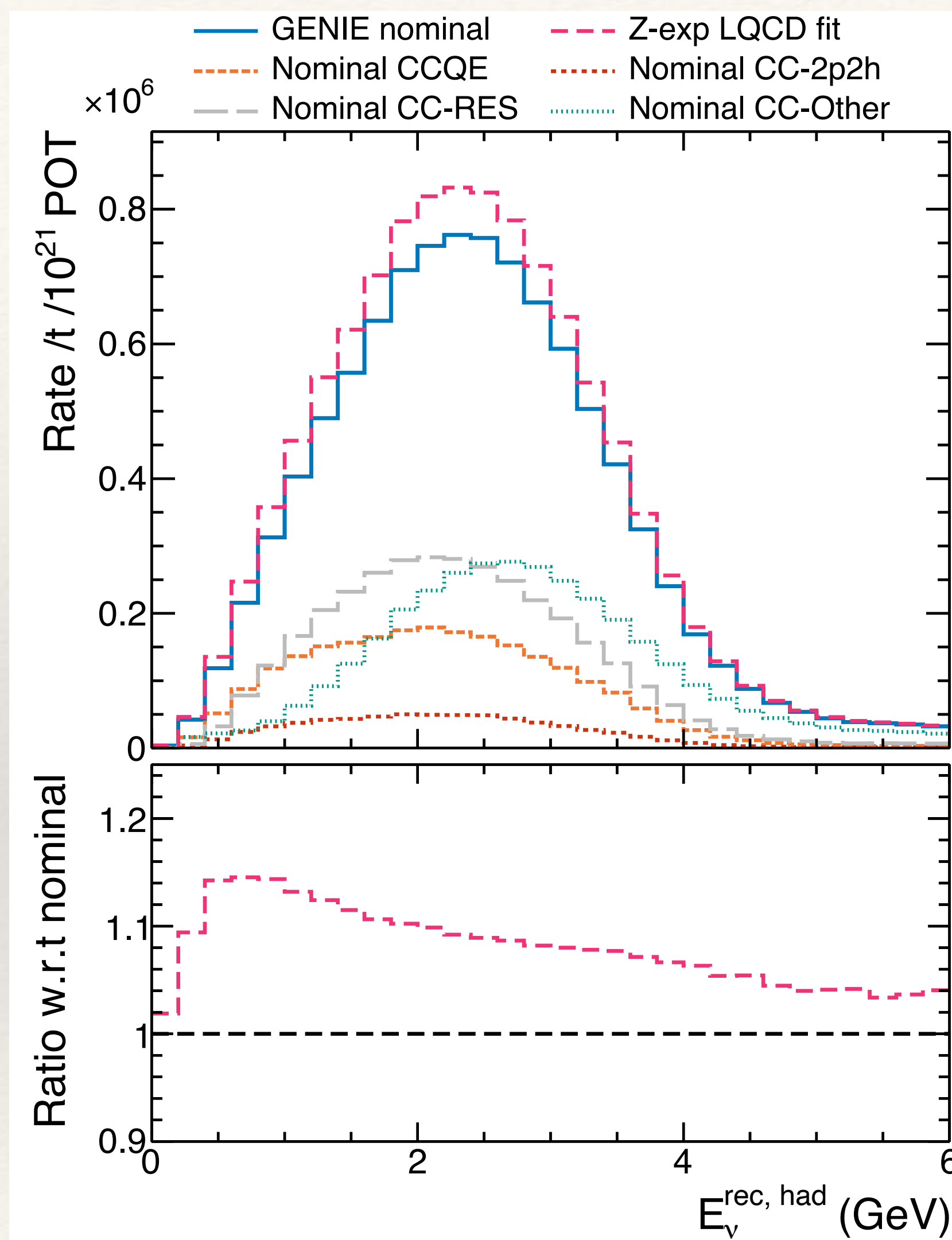
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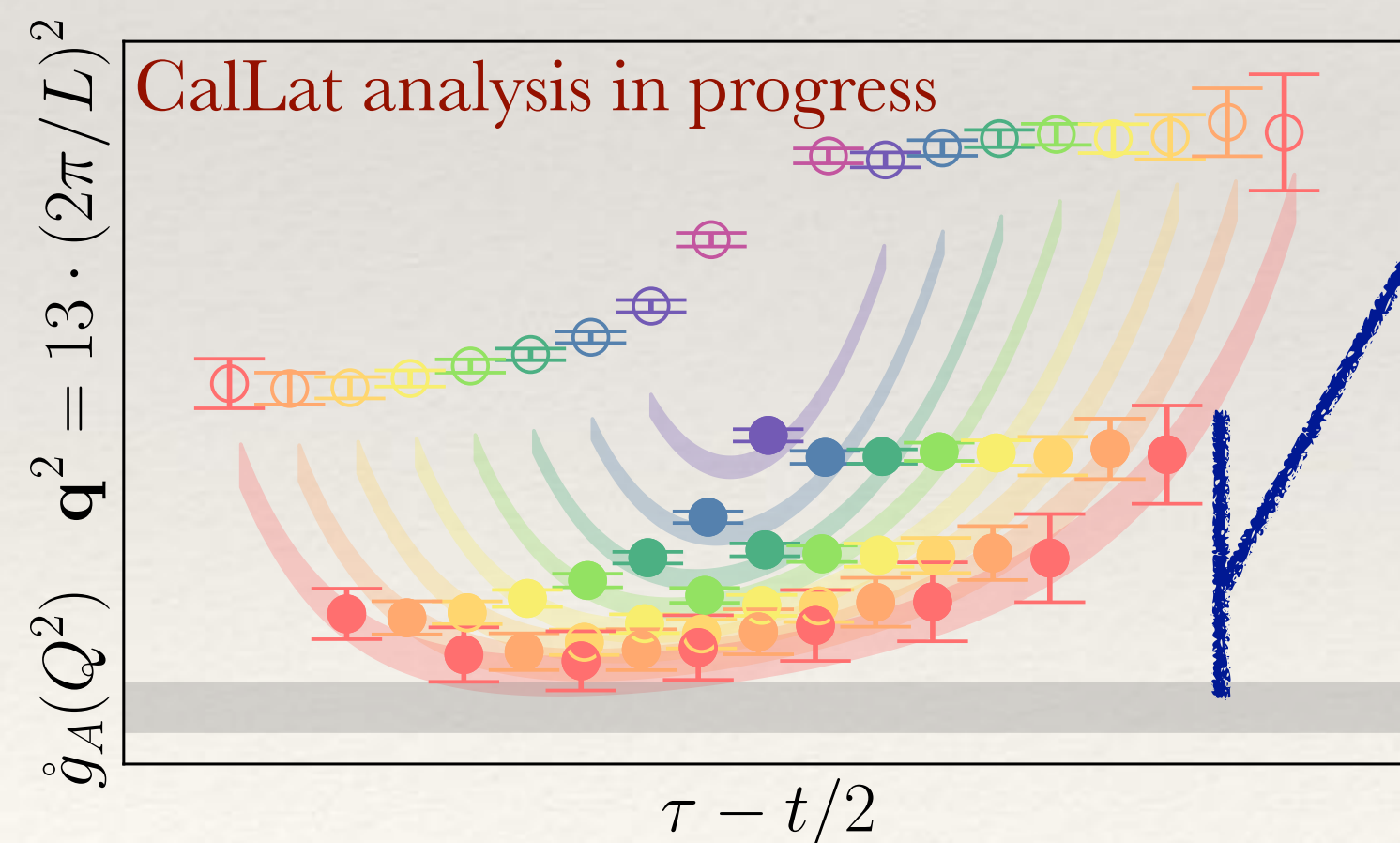
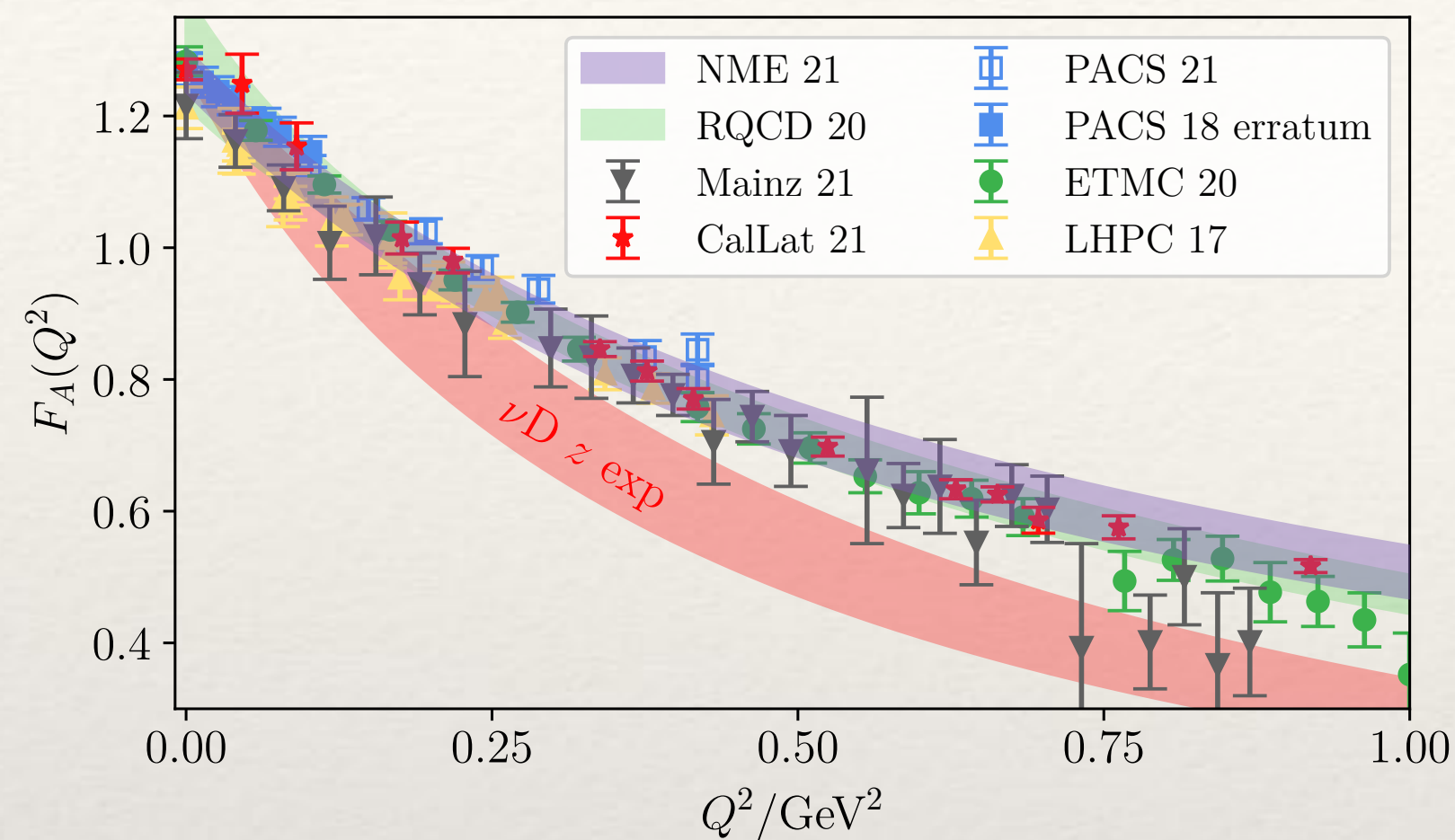
□ Energy dependent change

□ Different at near and far detector
— would interfere with ν energy reconstruction and determination of oscillation parameters



LQCD: Single Nucleon Axial Form Factor

Meyer, Walker-Loud, Wilkinson
Ann. Rev. Nucl. Part. Sci. 72 (2022)



❑ **What is required to finalize the lattice QCD results?**

❑ All lattice calculations are performed with the same strategy

❑ use different $t_{\text{sink}} - t_{\text{source}}$ separation to understand and model excited state contamination

$$\langle N(t_{\text{sink}}, \mathbf{p}_f = 0) | j_\mu^A(\tau, \mathbf{q}) | N(t_{\text{source}}, \mathbf{p}_i = -\mathbf{q}) \rangle$$

❑ No calculations are/can be performed with large enough $t_{\text{sink}} - t_{\text{source}}$ to ignore excited state contamination

❑ **We need a new strategy that has a different dependence on excited states**

❑ Enable momentum-space creation operators

$$\langle N(t_{\text{sink}}, \mathbf{p}_f) | j_\mu^A(\tau, \mathbf{q} = \mathbf{p}_f - \mathbf{p}_i) | N(t_{\text{source}}, \mathbf{p}_i) \rangle$$

❑ It also allows

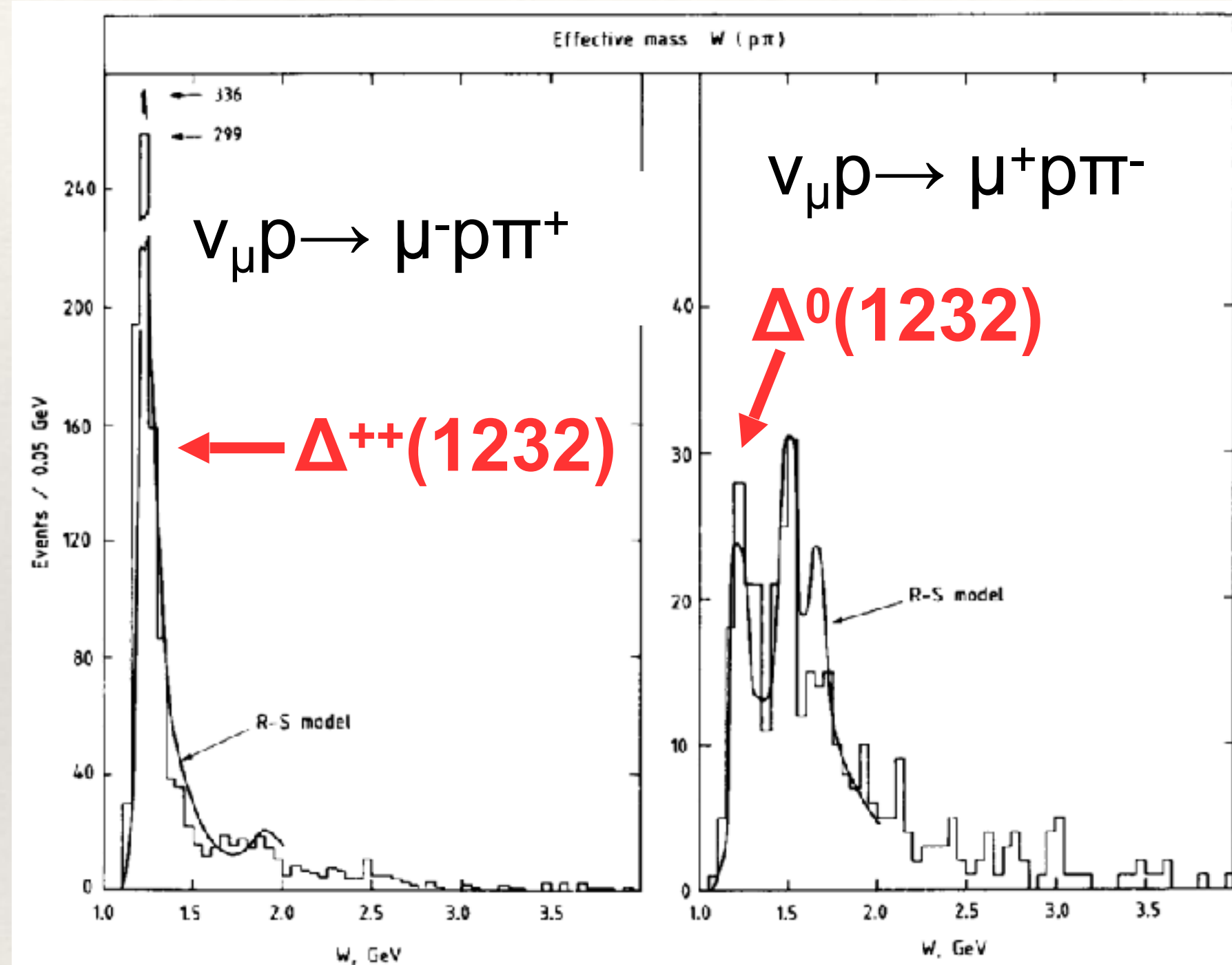
$$\langle \Delta | j_\mu | N \rangle, \quad \langle N\pi | j_\mu | N \rangle, \quad \langle NN | j_\mu | NN \rangle$$

❑ See Barca, Bali, Collins, PRD 107 (2023) [2211.12278]

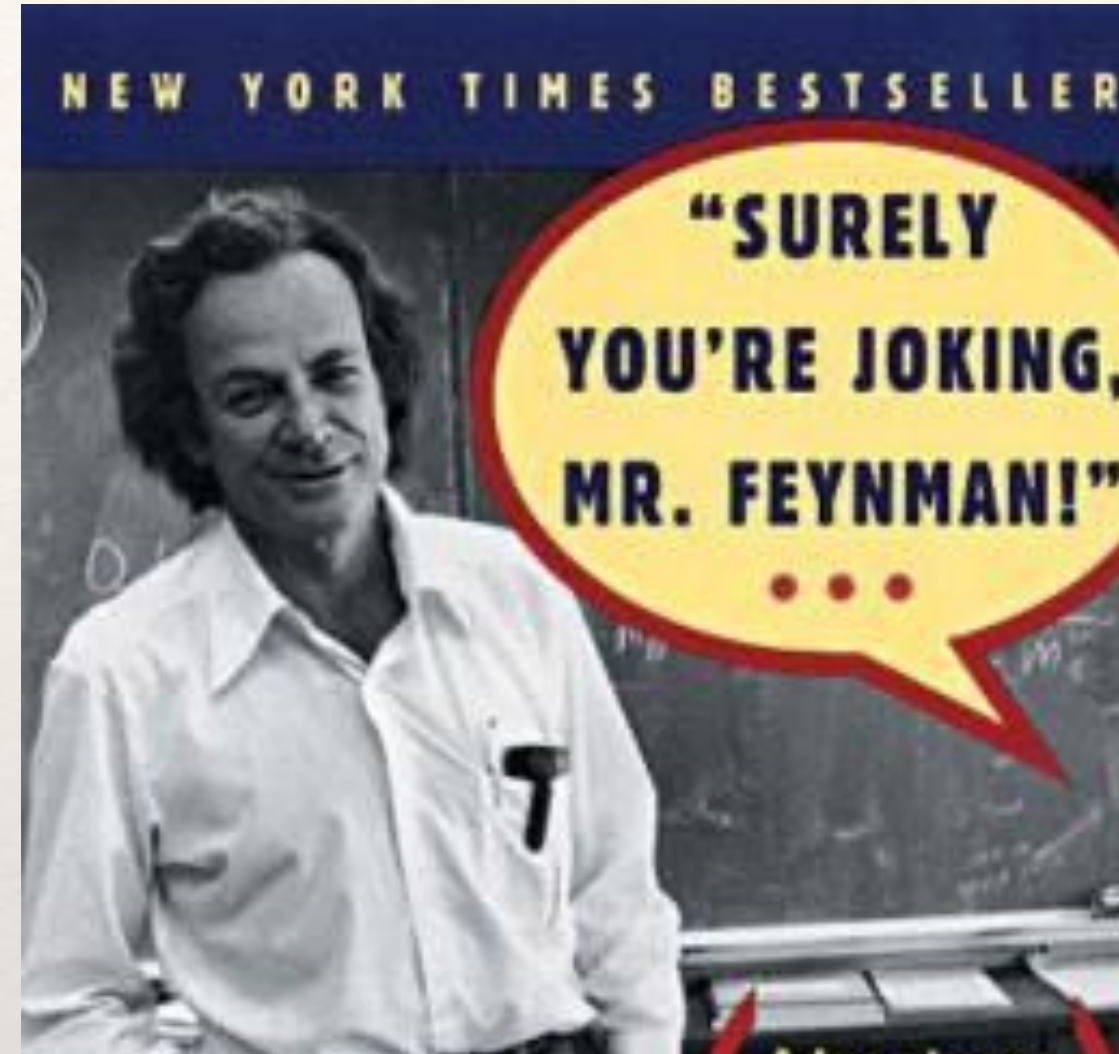
LQCD: Axial structure: next steps

State of the resonance Field

Nucl. Phys. B264 221 (1986)



$$W^2 = (\Sigma E)^2 - |\Sigma \mathbf{p}|^2$$



Indeed not!

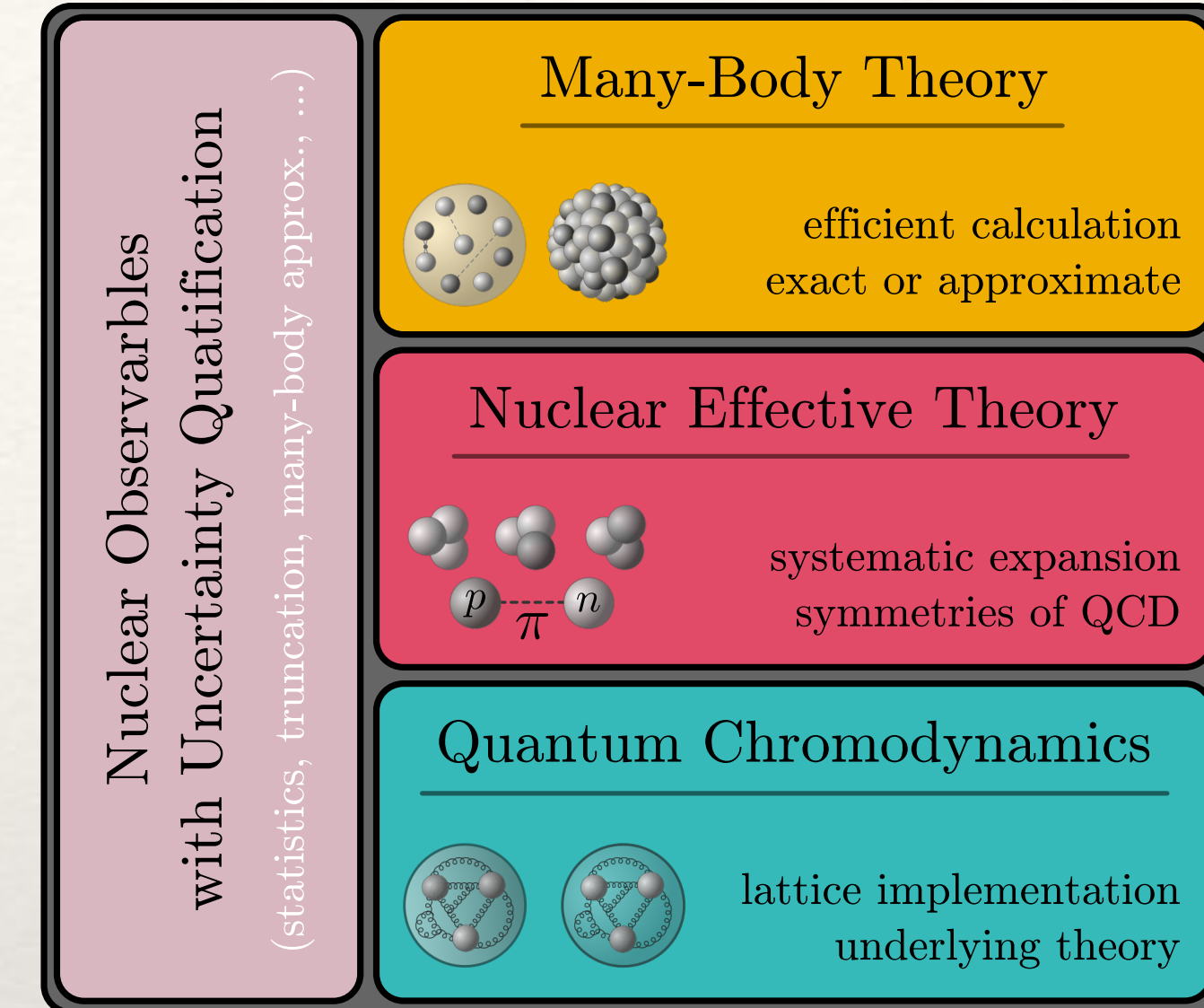
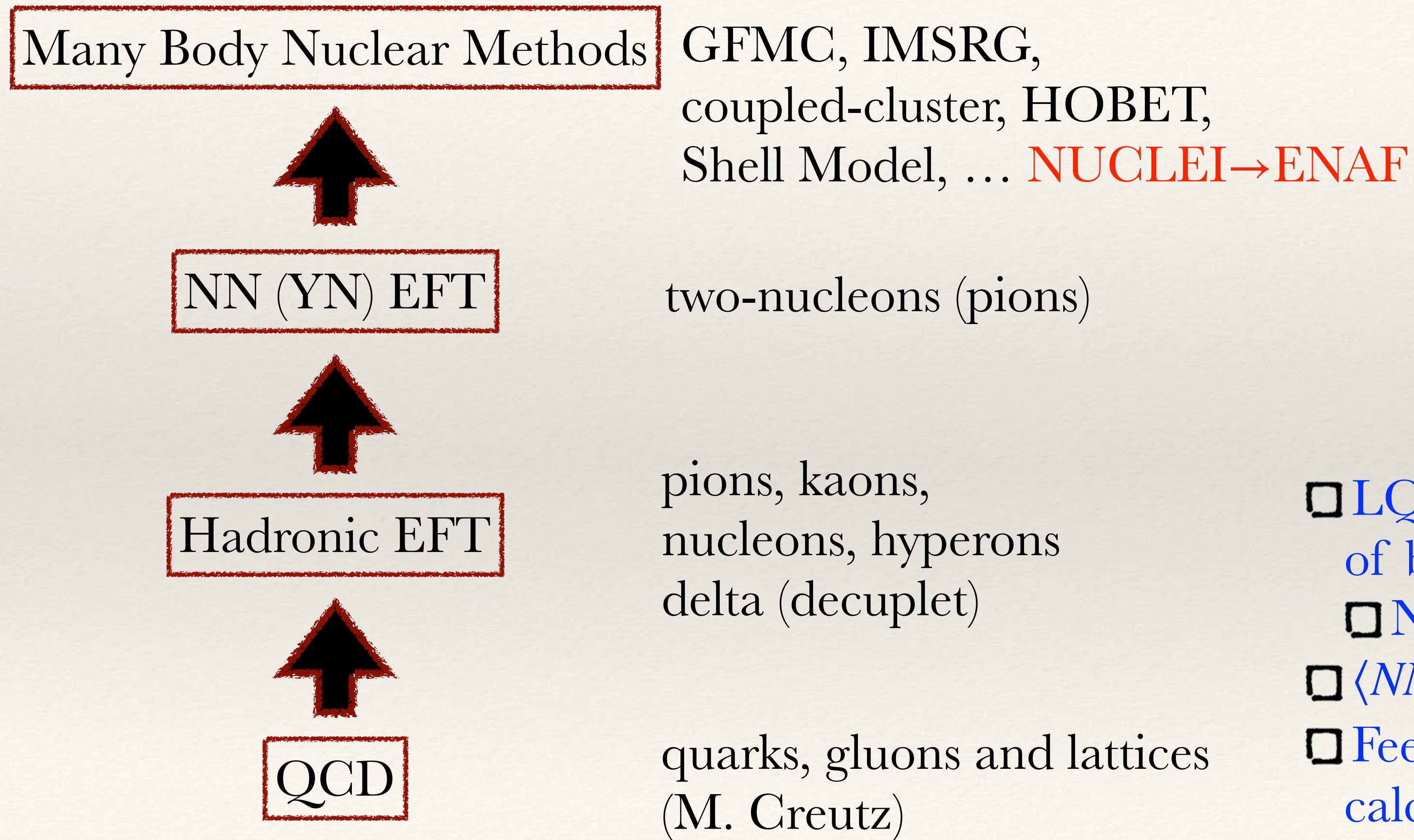
Our pion production model uses a description of resonance production that is “naive and obviously wrong in its simplicity” [Feynman, Kislinger, Ravndal PRD3 (1971)]

“I trust some bright motivated physicists will fix this soon”

- ☐ Current models are unsatisfactory:
 - ☐ Simplistic description of neutrino-nucleon interaction
 - ☐ Unsophisticated description of the nucleus
- ☐ Heavy reliance on old data (experiments shut down)
- ☐ ~10% uncertainties on effective parameters at best
- ☐ insufficient for precision goals of DUNE (C. Wilkinson, private communication)

LQCD and EFT for $\nu - A$ (and other interesting NP problems)

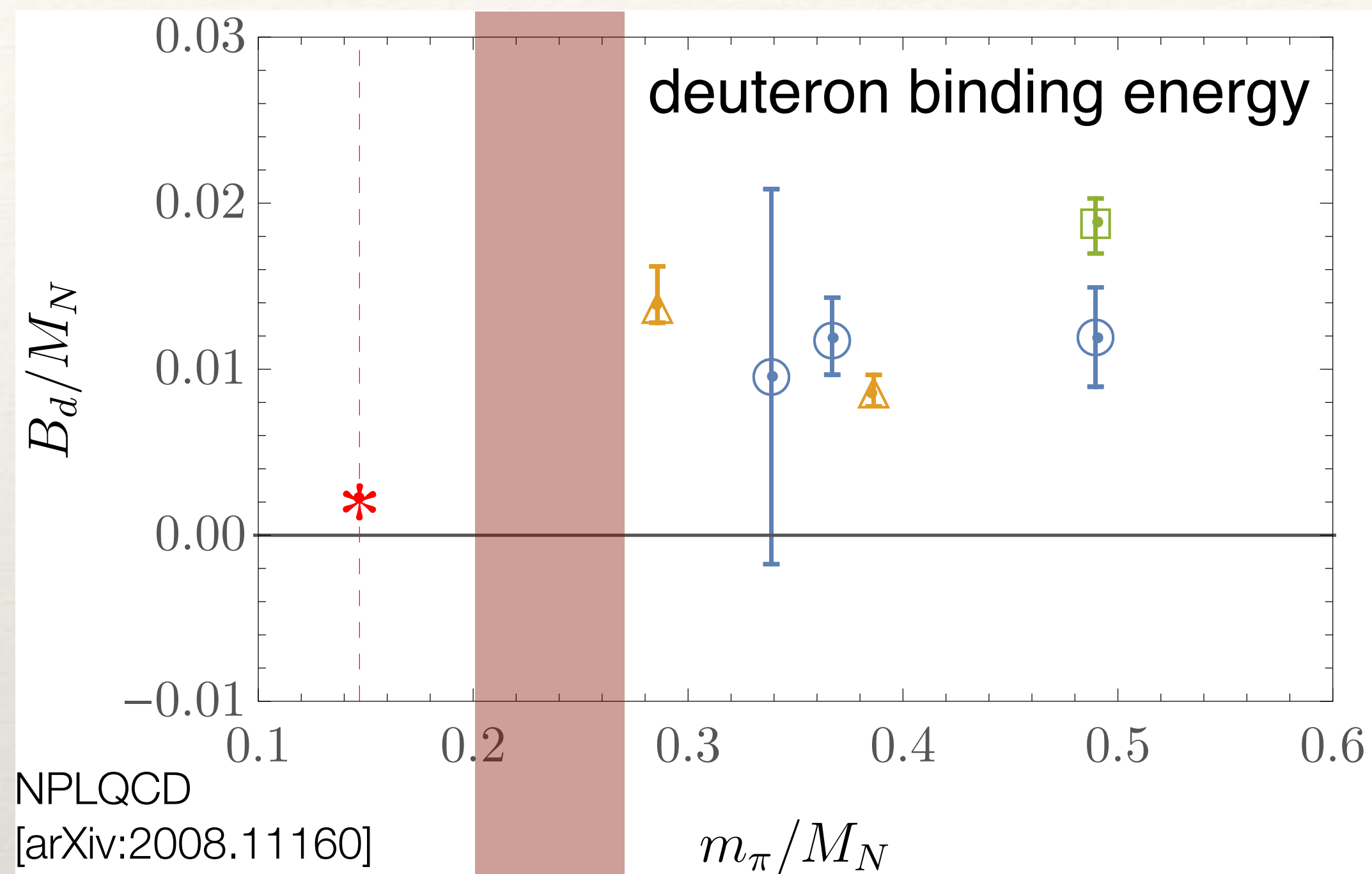
- Quantitatively understand the emergence of nuclear physics from the Standard Model (SM) and predict matrix elements needed for BSM searches



(fig: C. Drescher)

- LQCD to map out convergence of baryon XPT (Δ)
- Need calculations w/ Δ
- $\langle NN | j_\mu | NN \rangle$
- Feed results into many-body NP calculations — **NUCLEI**

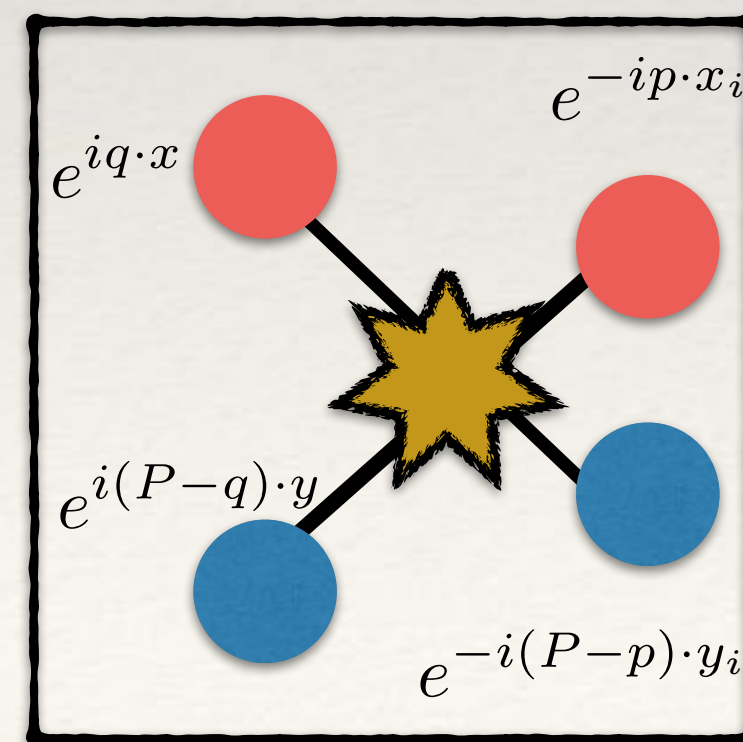
Survey of lattice QCD results for two-baryons



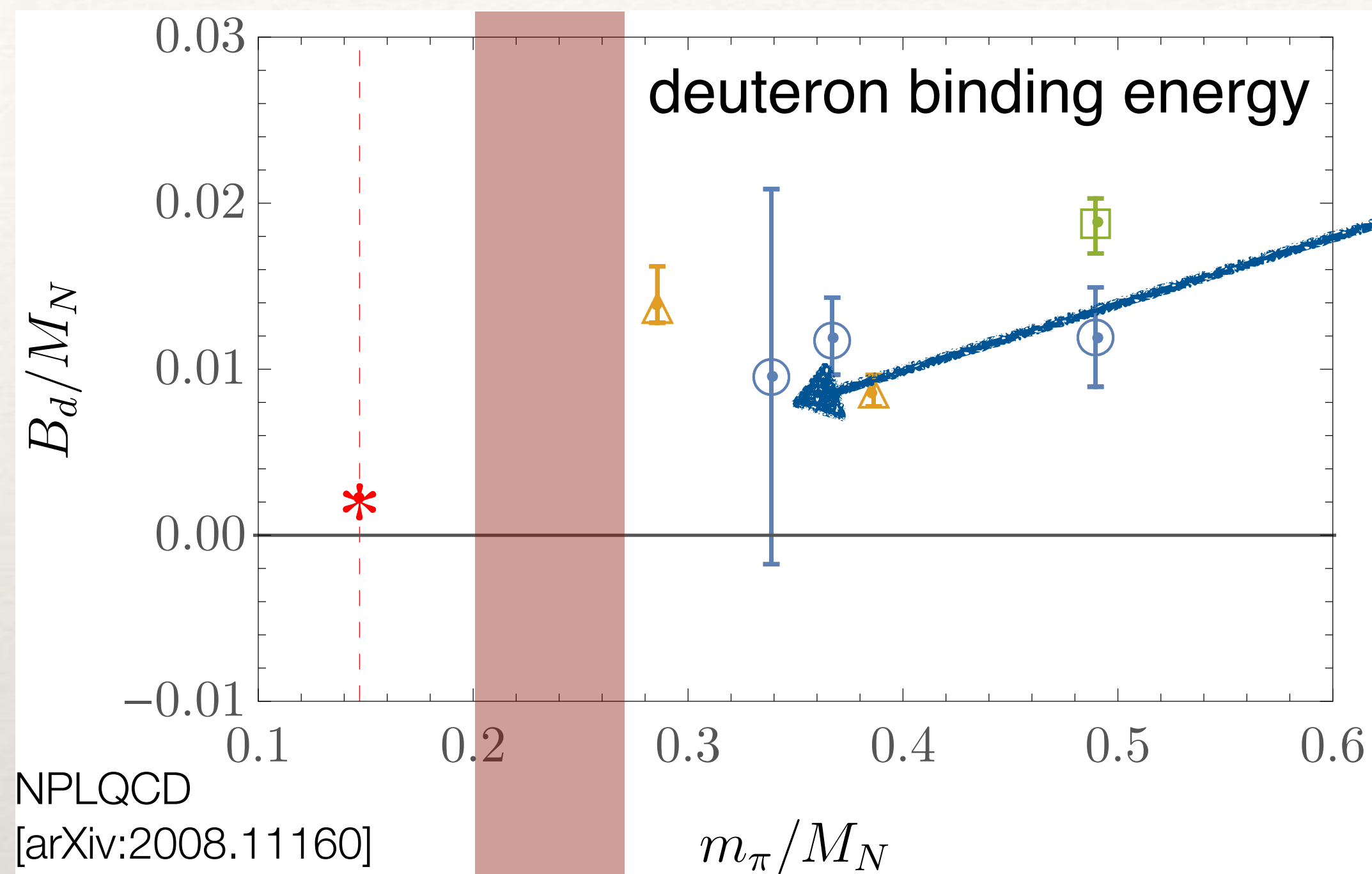
LQCD Results with (deeply) bound di-nucleons

2006 NPLQCD - first dynamical LQCD calculations of NN

Estimated upper range of
validity of NN EFT



Survey of lattice QCD results for two-baryons



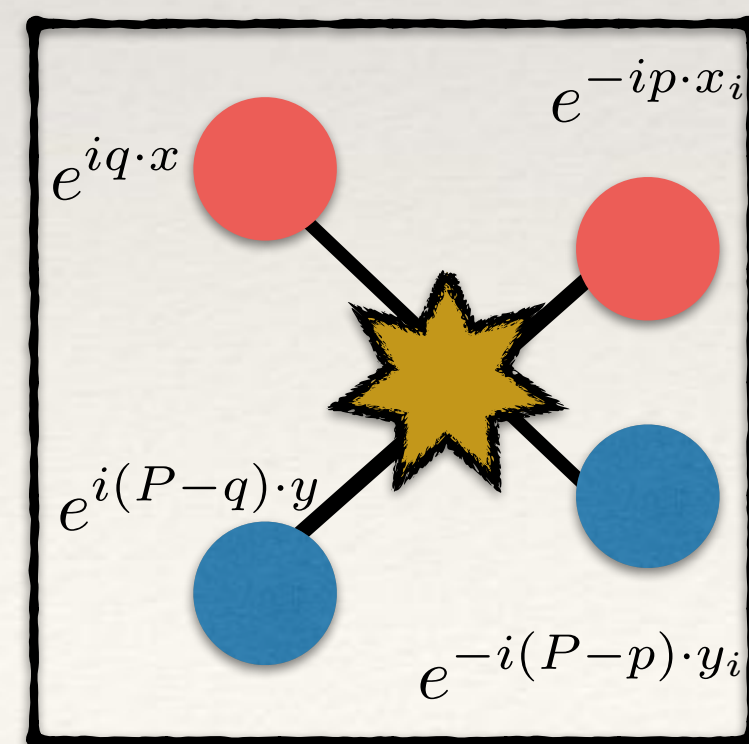
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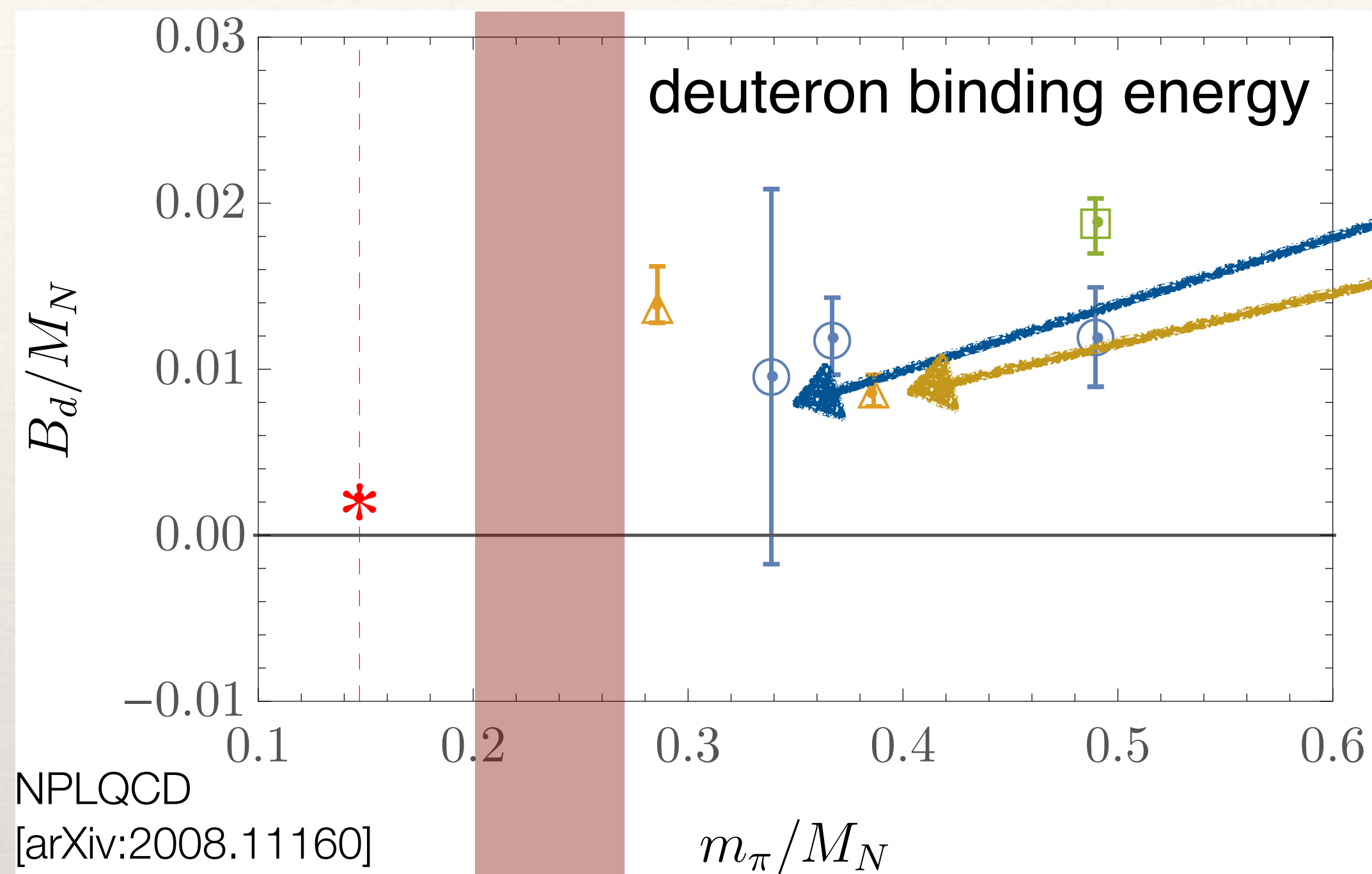
2011 NPLQCD

$M_\pi \approx 390$ MeV

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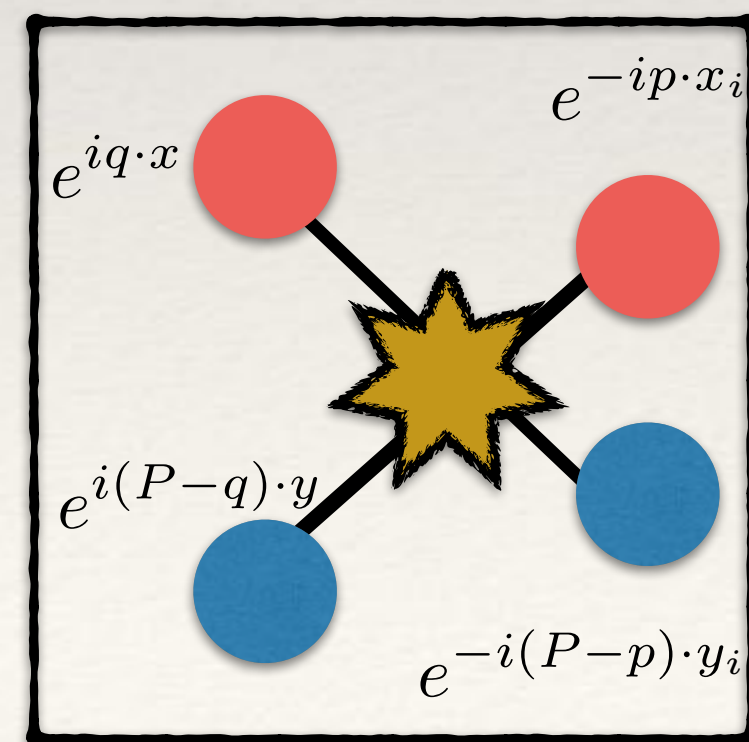
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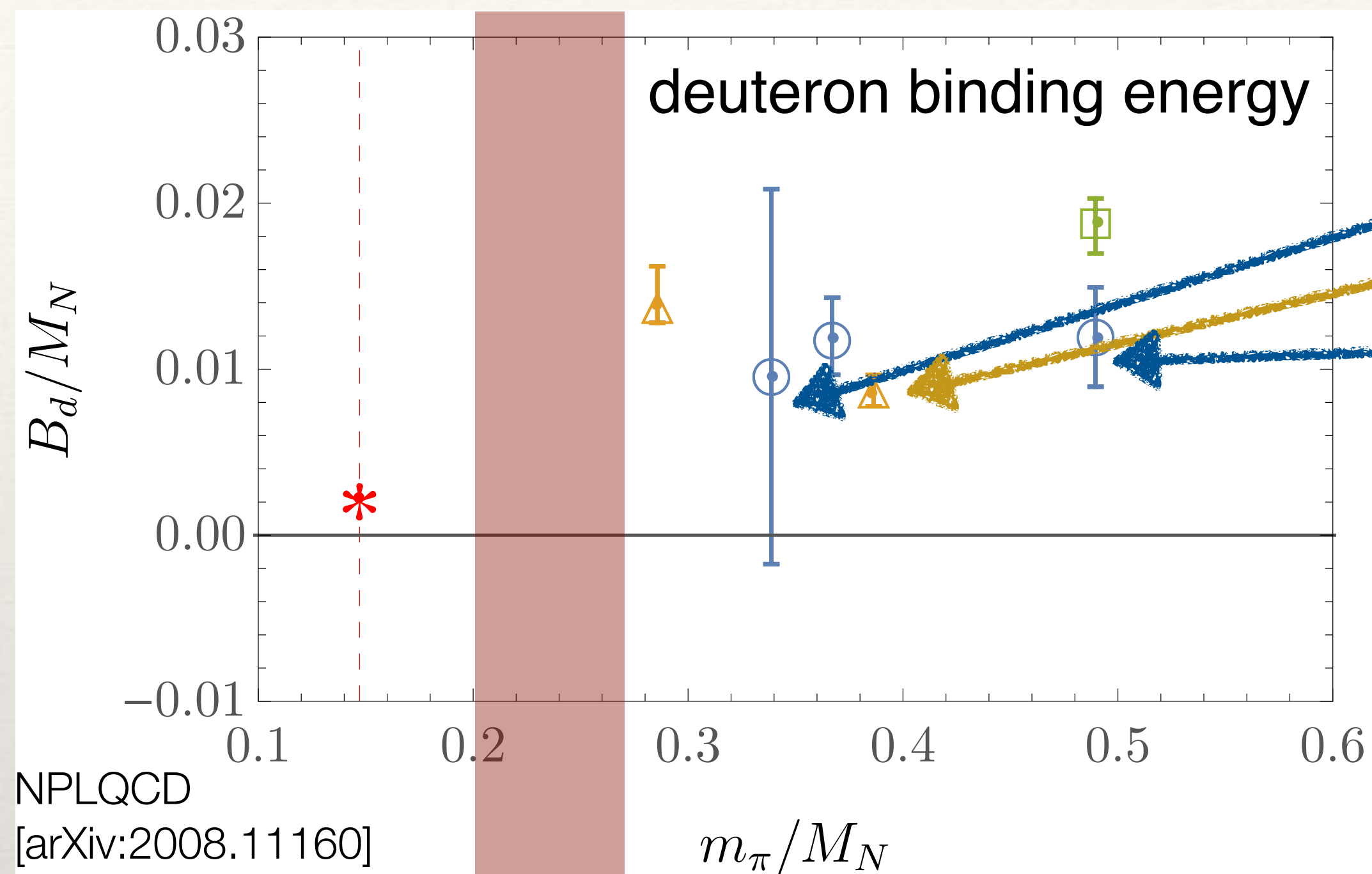
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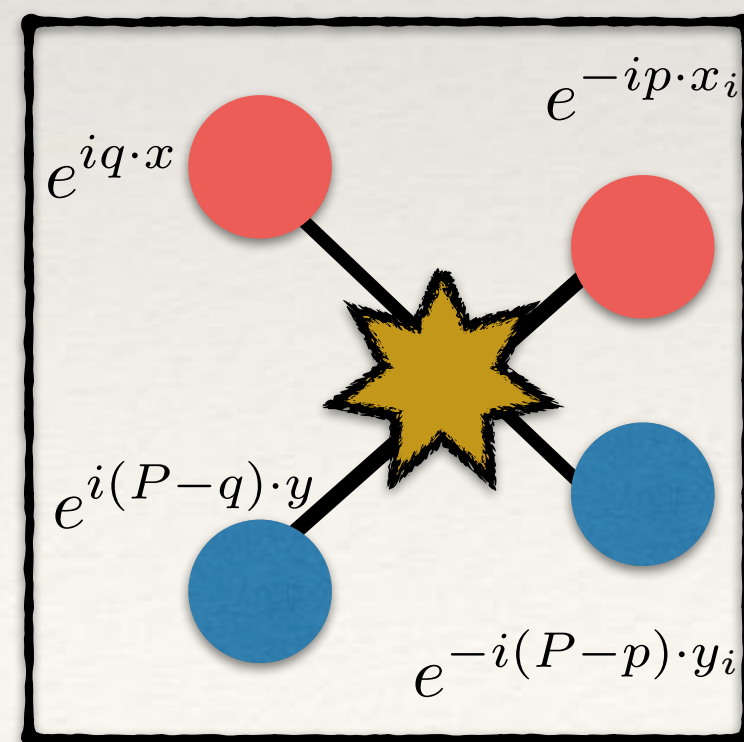
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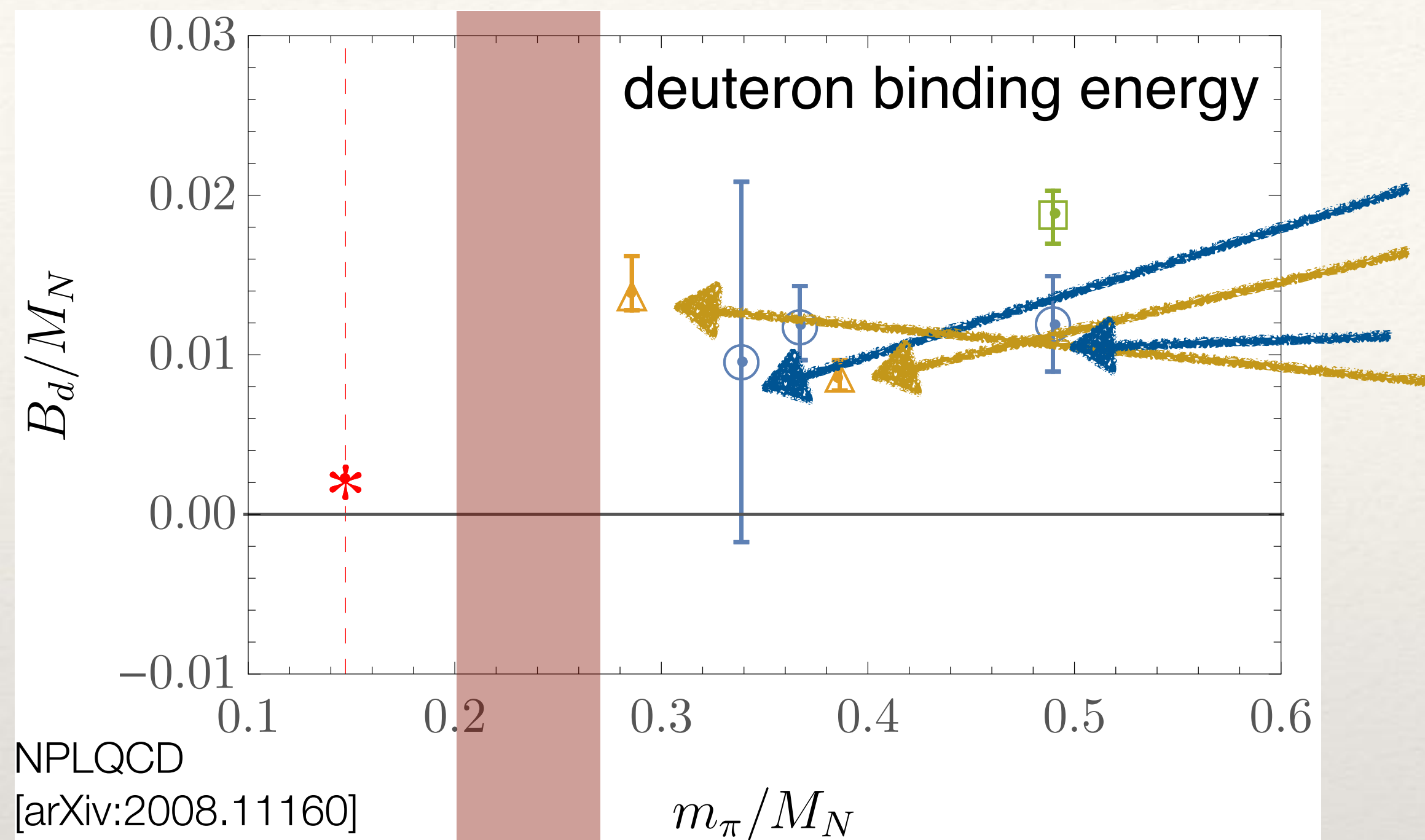
2012 Yamazaki et al. $M_\pi \approx 510$ MeV

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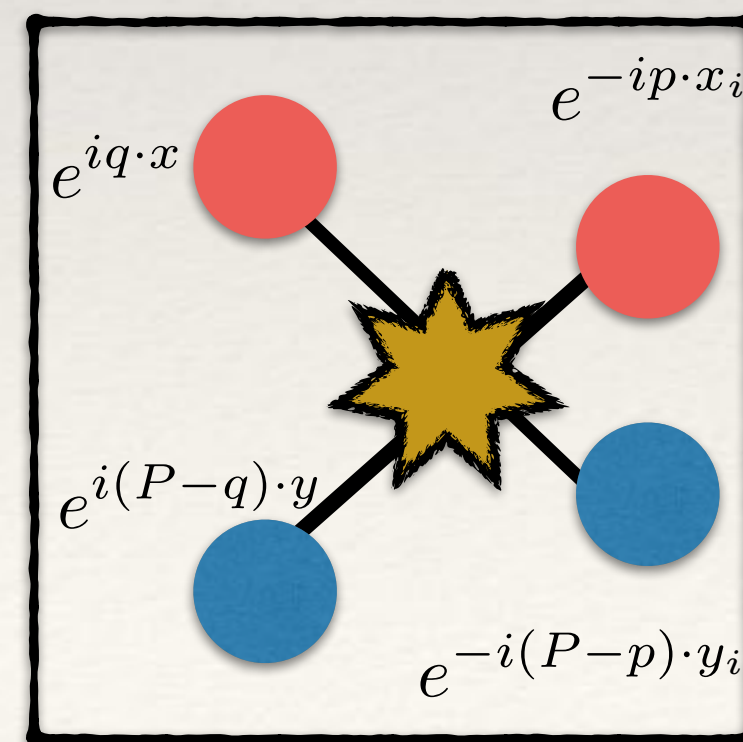
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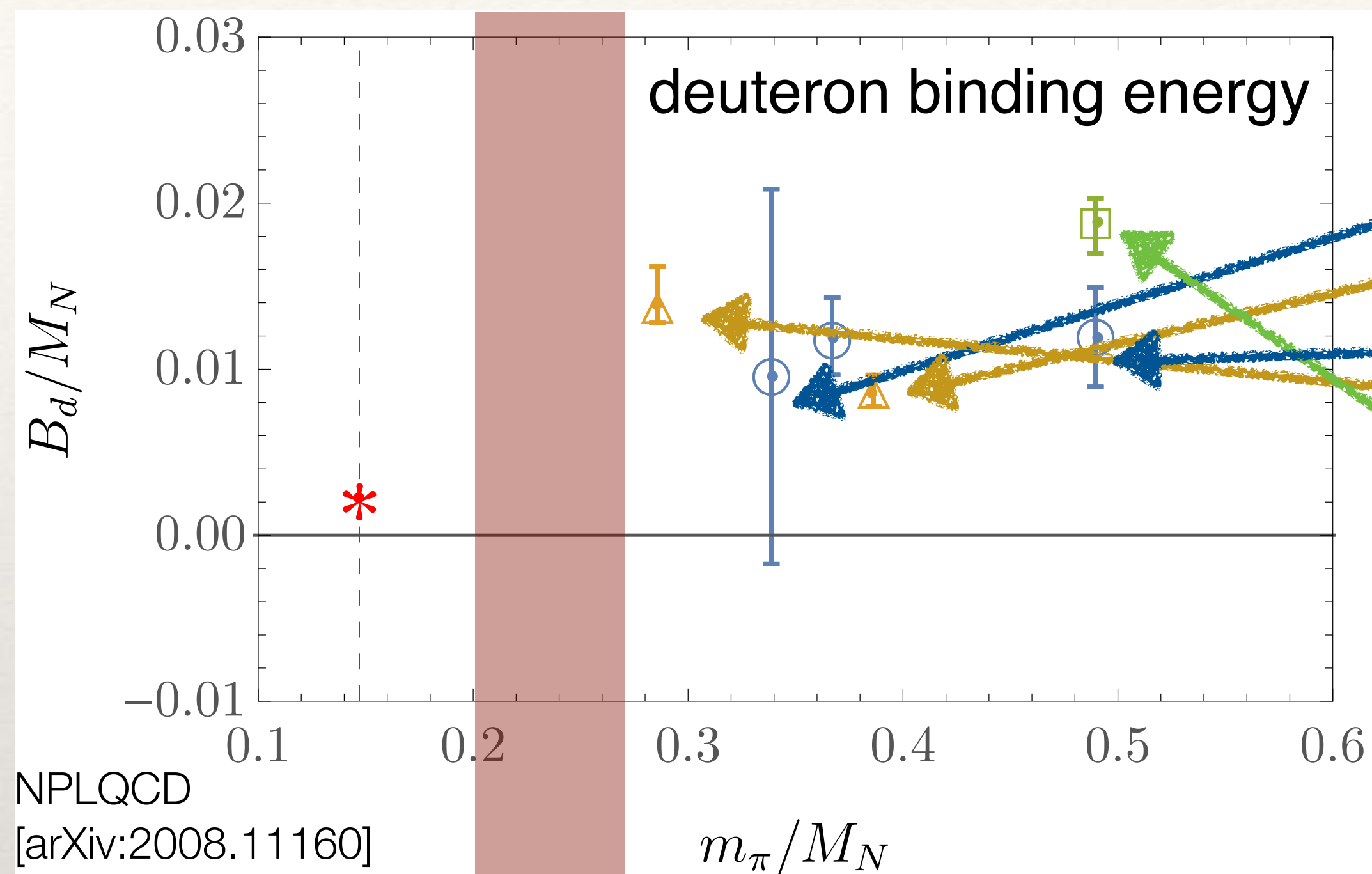
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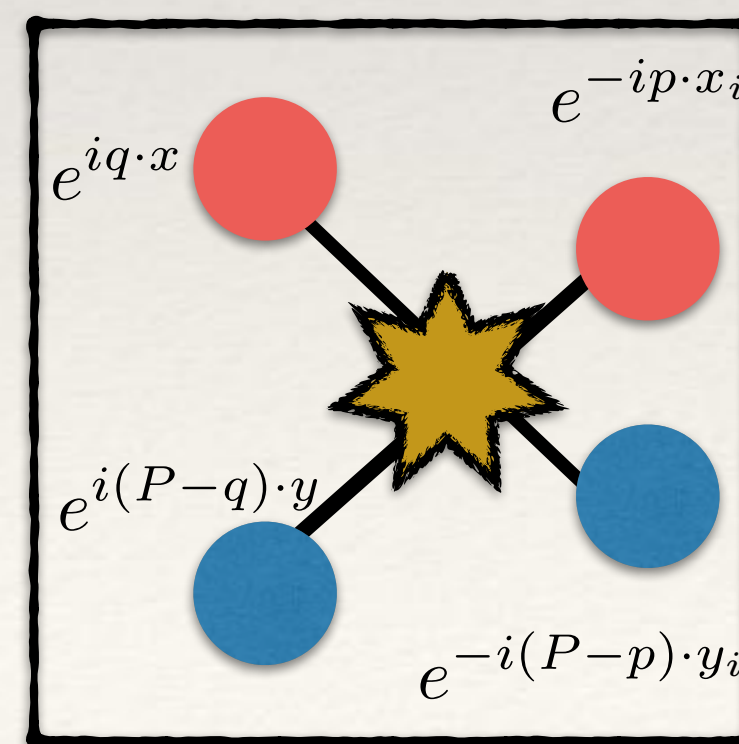
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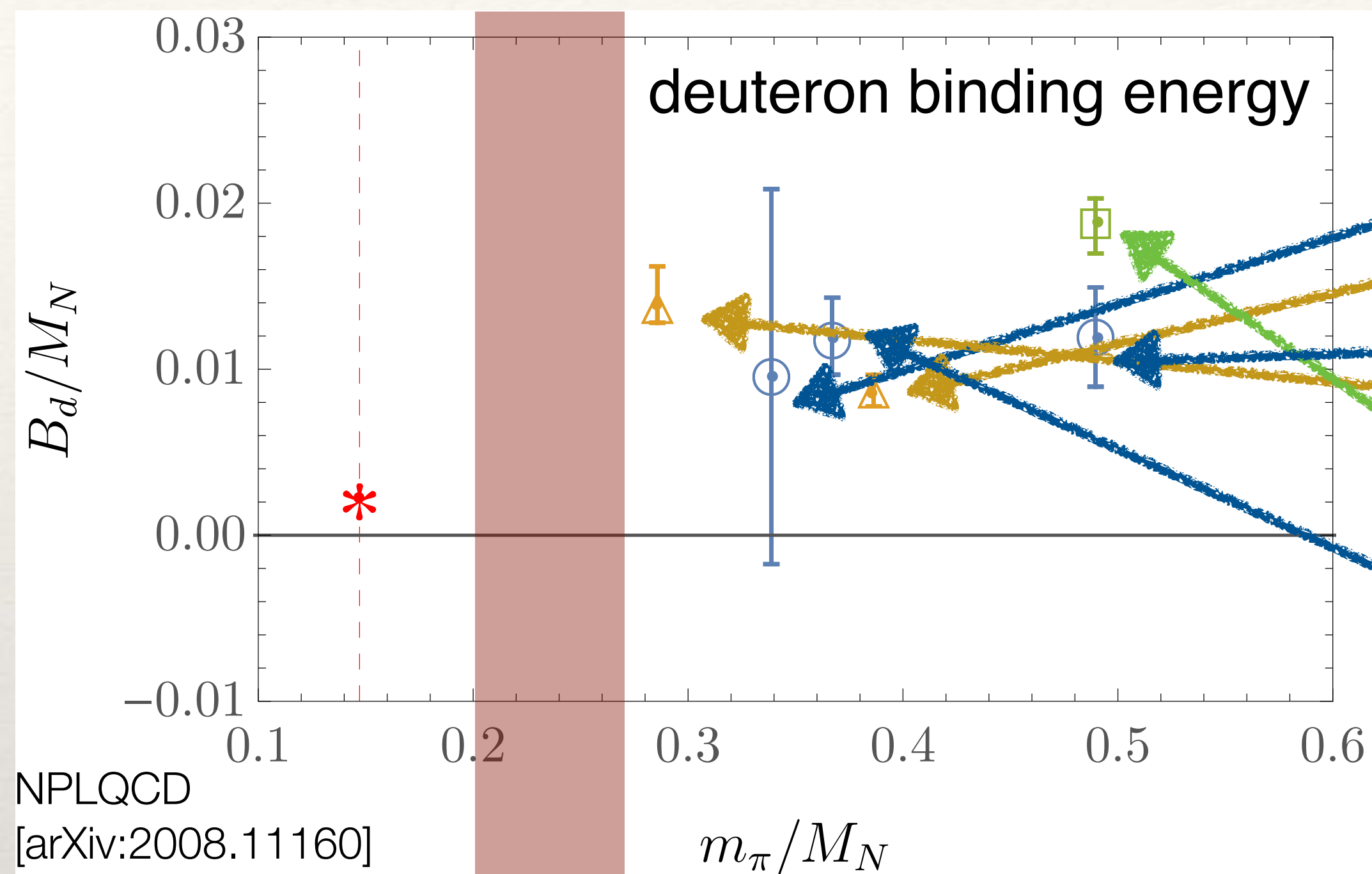
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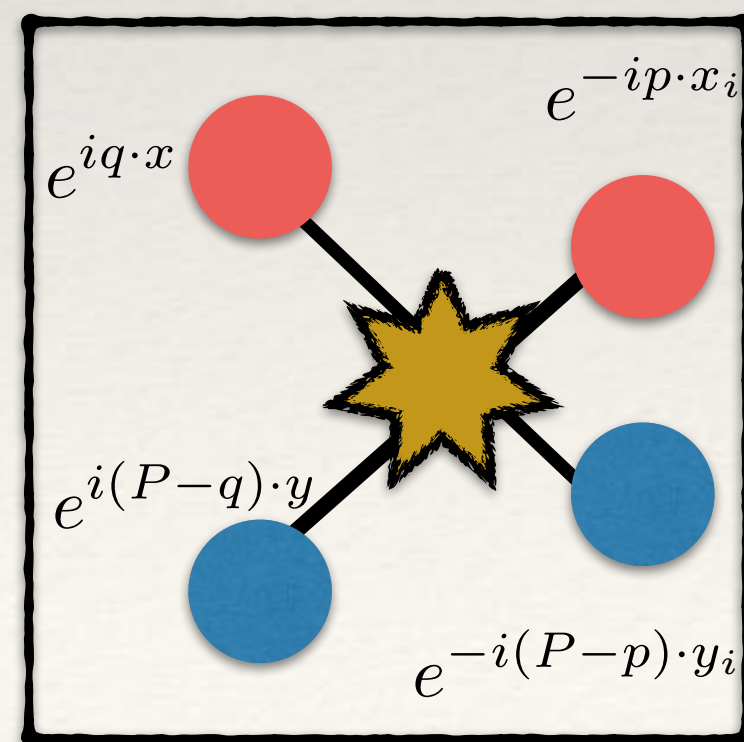
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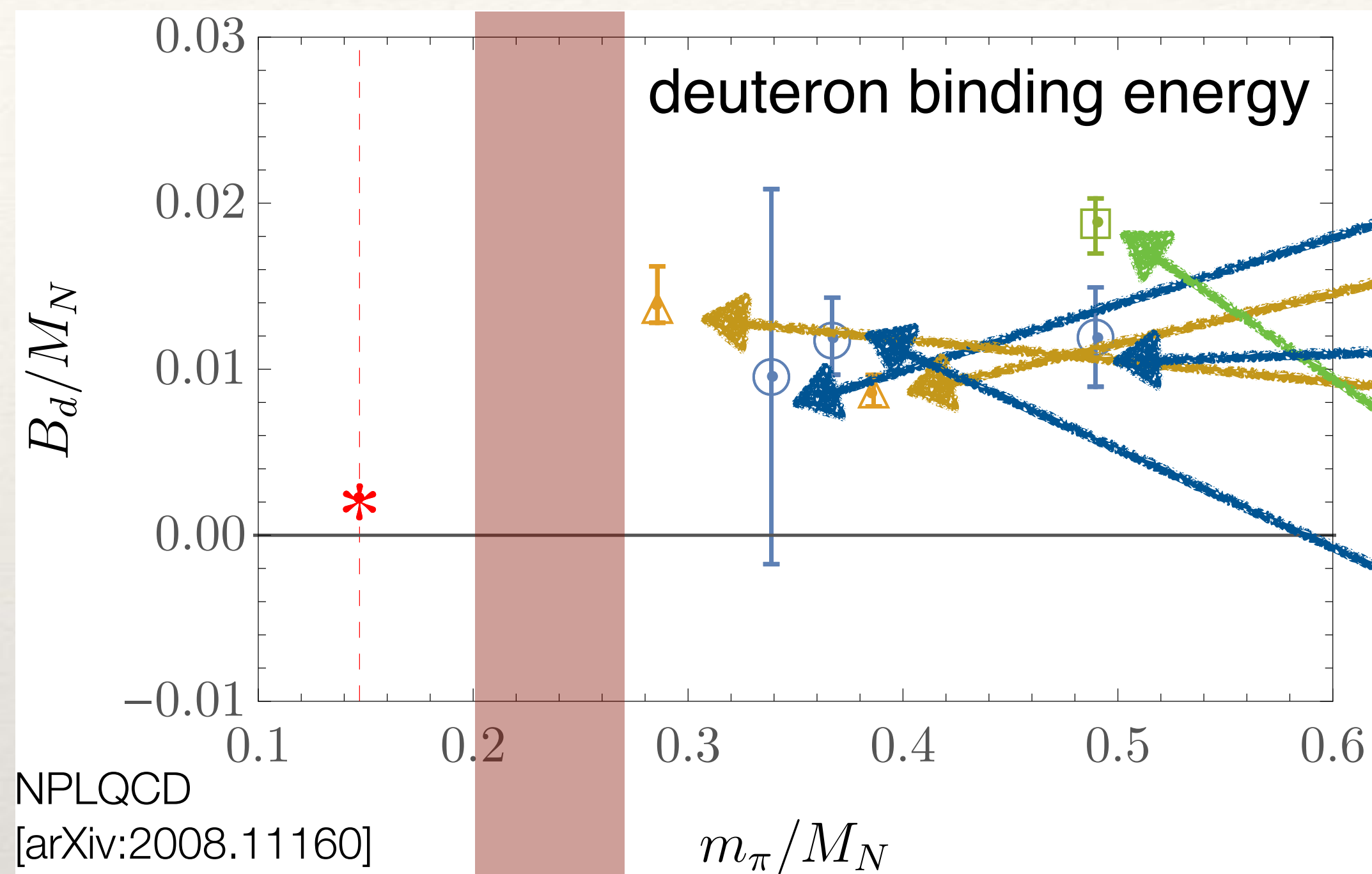
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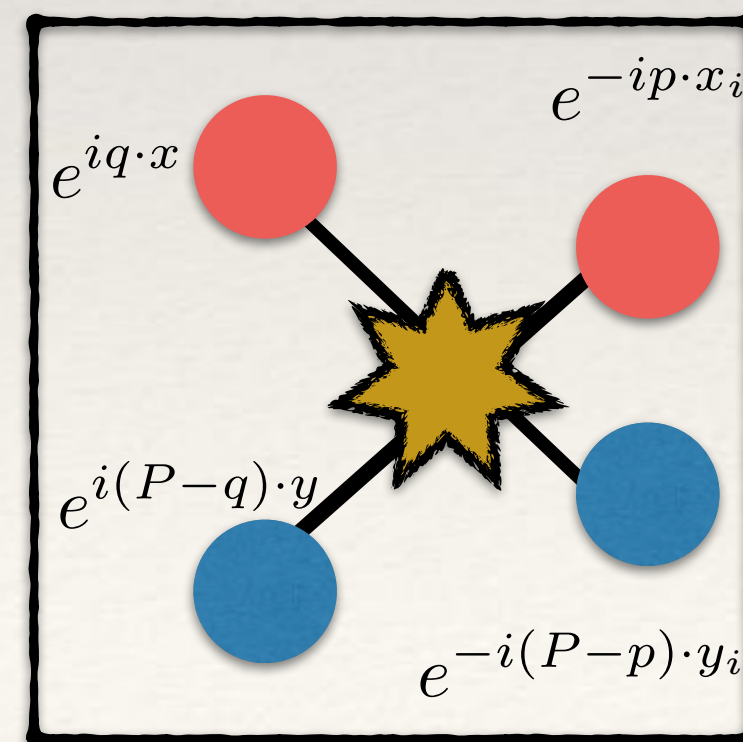
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LQCD Results without bound di-nucleons (or inconclusive)

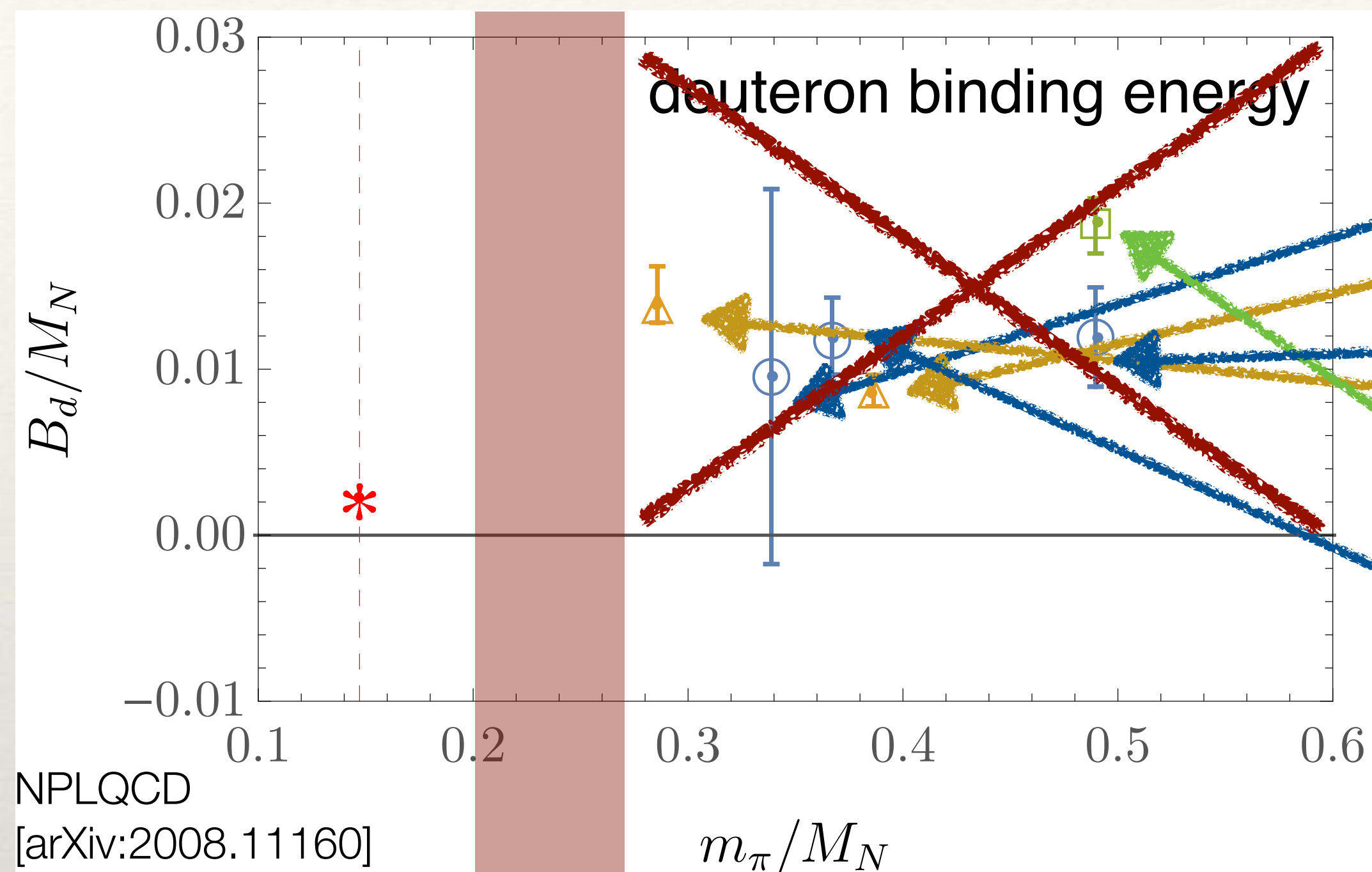
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2020	CoSMoN	$M_\pi \approx 714$ MeV
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Estimated upper range of
validity of NN EFT



(blue = work I was involved in)

Survey of lattice QCD results for two-baryons



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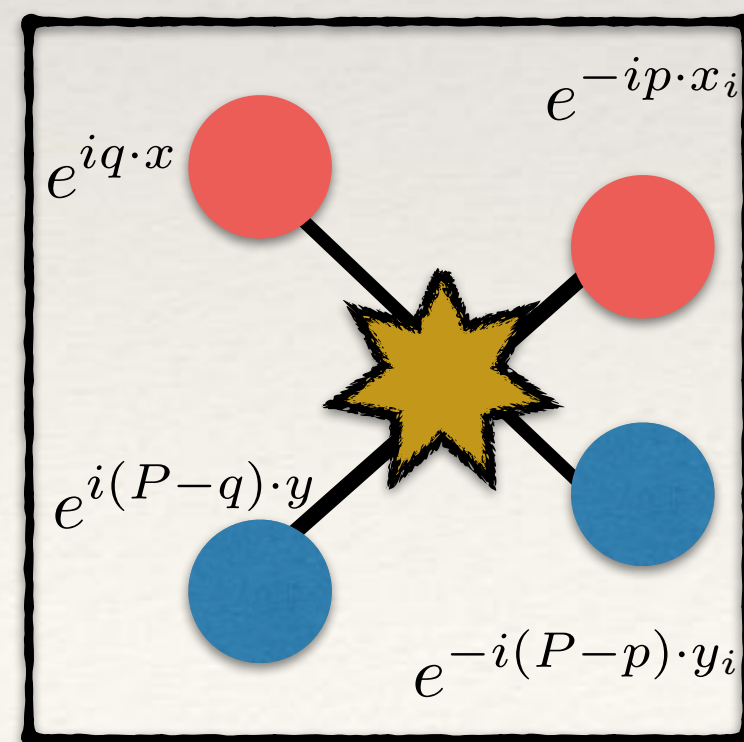
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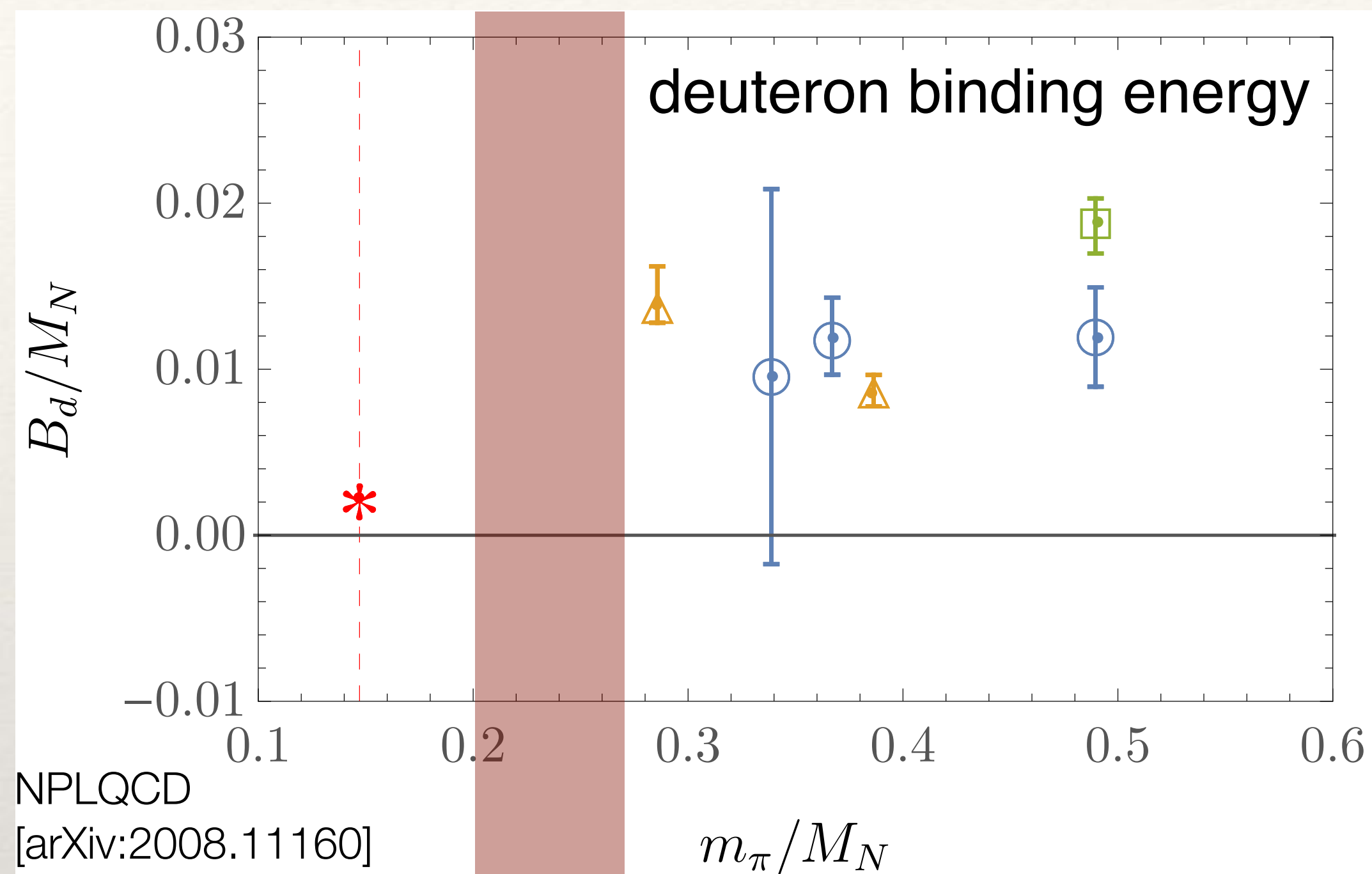
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**We now believe the bound state
results are not correct**



(blue = work I was involved in)

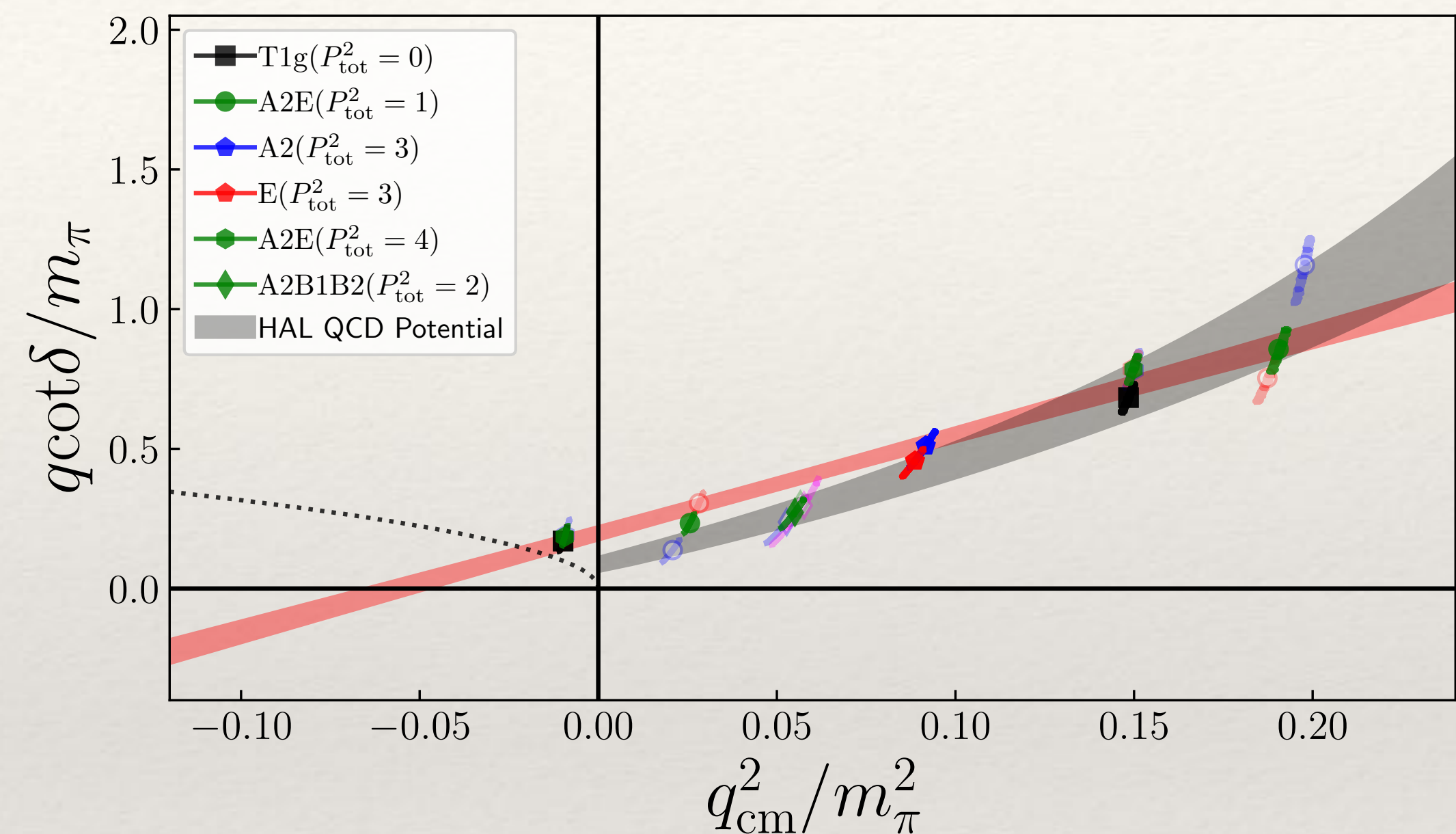
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Di-nucleons do not form bound states at heavy pion mass

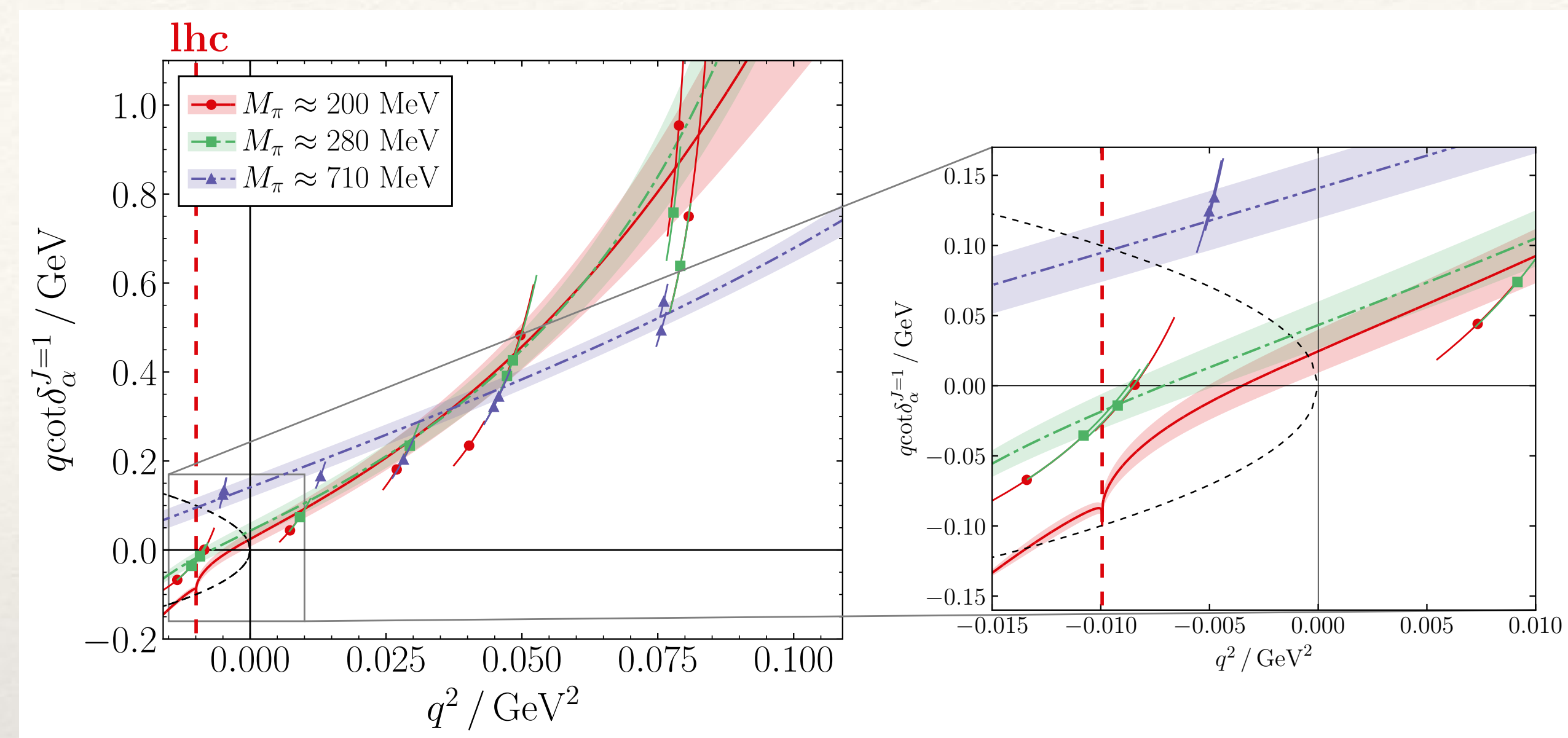
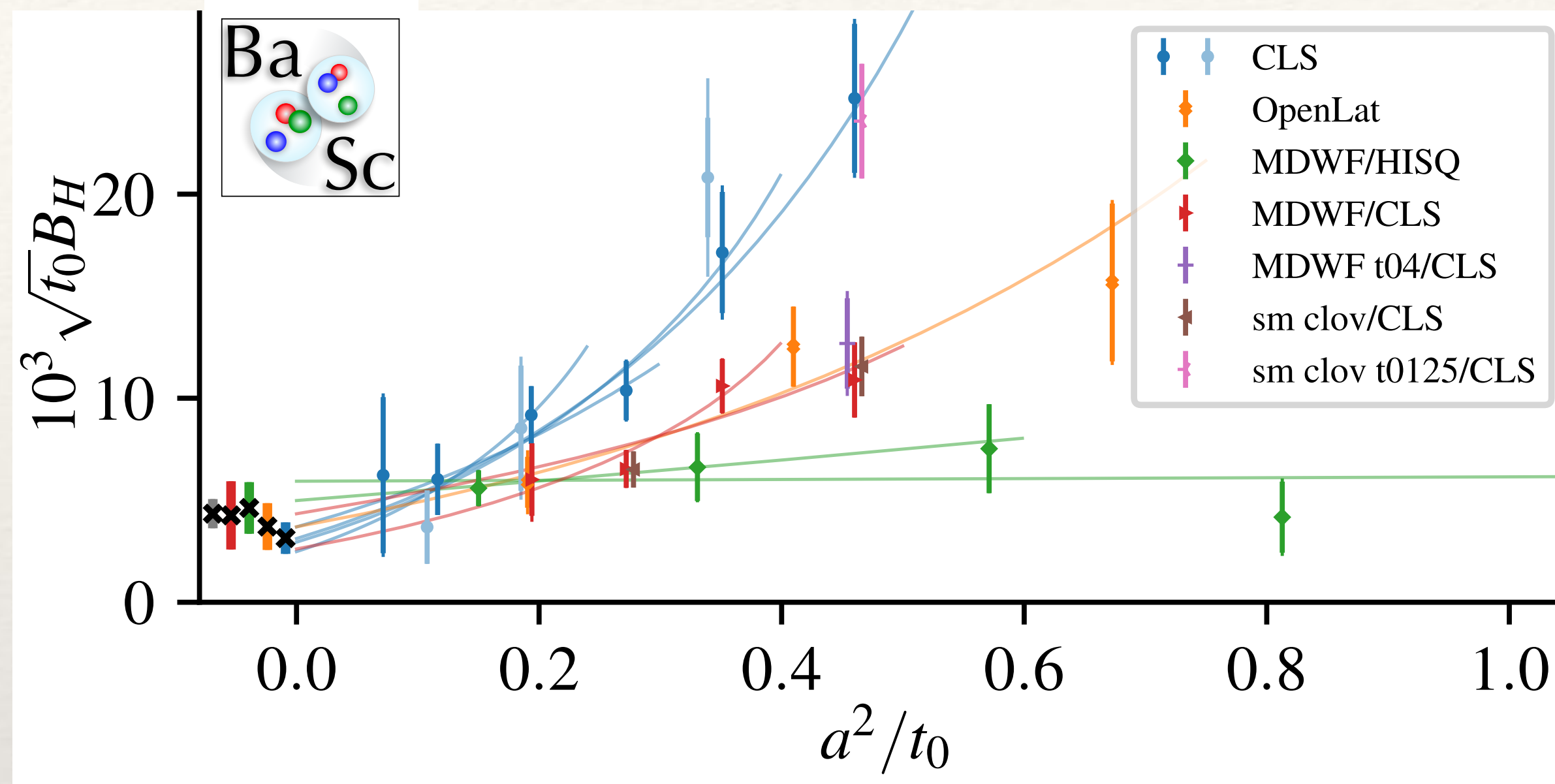
J. Bulava, ..., A. Walker-Loud,

Phys.Rev.C 113 (2026) 2, 024002 [arXiv: 2505.05547] *Editor's Suggestion*



- Resolves 15 year old discrepancy in literature
- Shows equivalence between modern LQCD methods for di-nucleons and HAL QCD Potential method
- Excludes a bound di-nucleon in the deuteron channel by $\sim 7\sigma$
- Paves the way for di-nucleon calculations at lighter pion mass ($m_\pi \lesssim 280$ MeV)

Lattice QCD results for di-nucleons: going forward



❑ Large discretization errors observed in h-dibaryon binding energy LQCD calculations

❑ Very dependent upon quark-action

❑ MDWF - quark-action that preserves chiral symmetry (to high degree) has small corrections

❑ In preparation

❑ We are finalizing results with lighter pion mass points: $m_\pi \approx \{200, 280, 714\}$ MeV

❑ At what m_π does the deuteron bind?

❑ First quantitative comparison to NN EFT!
→ Connect with NUCLEI SciDAC

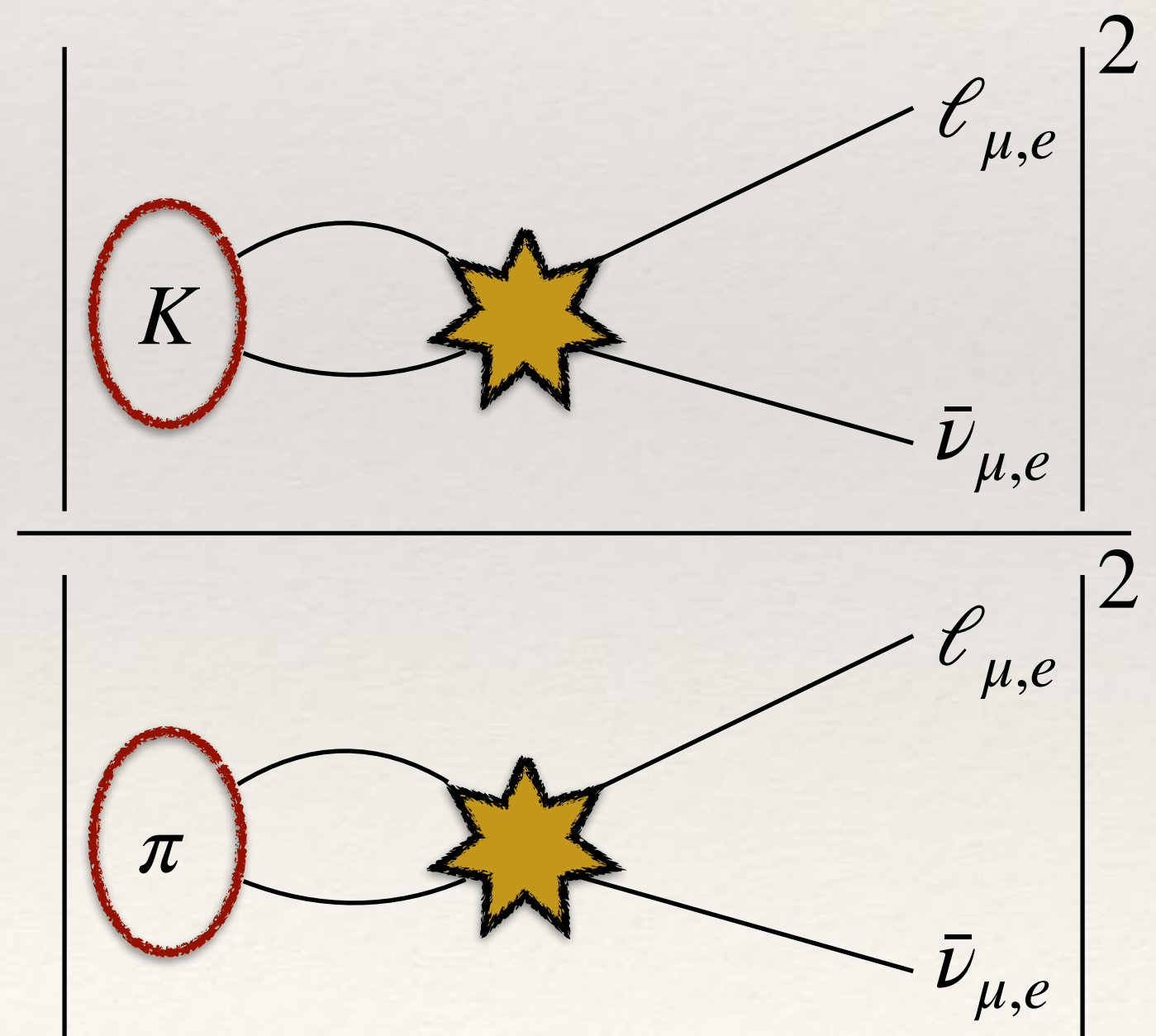
❑ In preparation

MDWF are more expensive (5 dimensional) — but really?

- ❑ When all costs taken in aggregate, are MDWF fermions really more expensive for nuclear physics applications?
- ❑ MDWF have smaller discretization errors: fewer and coarser lattice spacings for continuum extrapolation
- ❑ MDWF are more compute intense: better load balancing of computations on GPU machines and more wall clock time spent in highly optimized fermion solves (Grid, QUDA)
- ❑ MDWF are observed to have smaller statistical errors than clover: fewer total computations needed
- ❑ MDWF are simpler to renormalize: less postdoc years to carry out arduous renormalization, focus on physics instead
- ❑ My effort in this SciDAC is focussed on answering this question

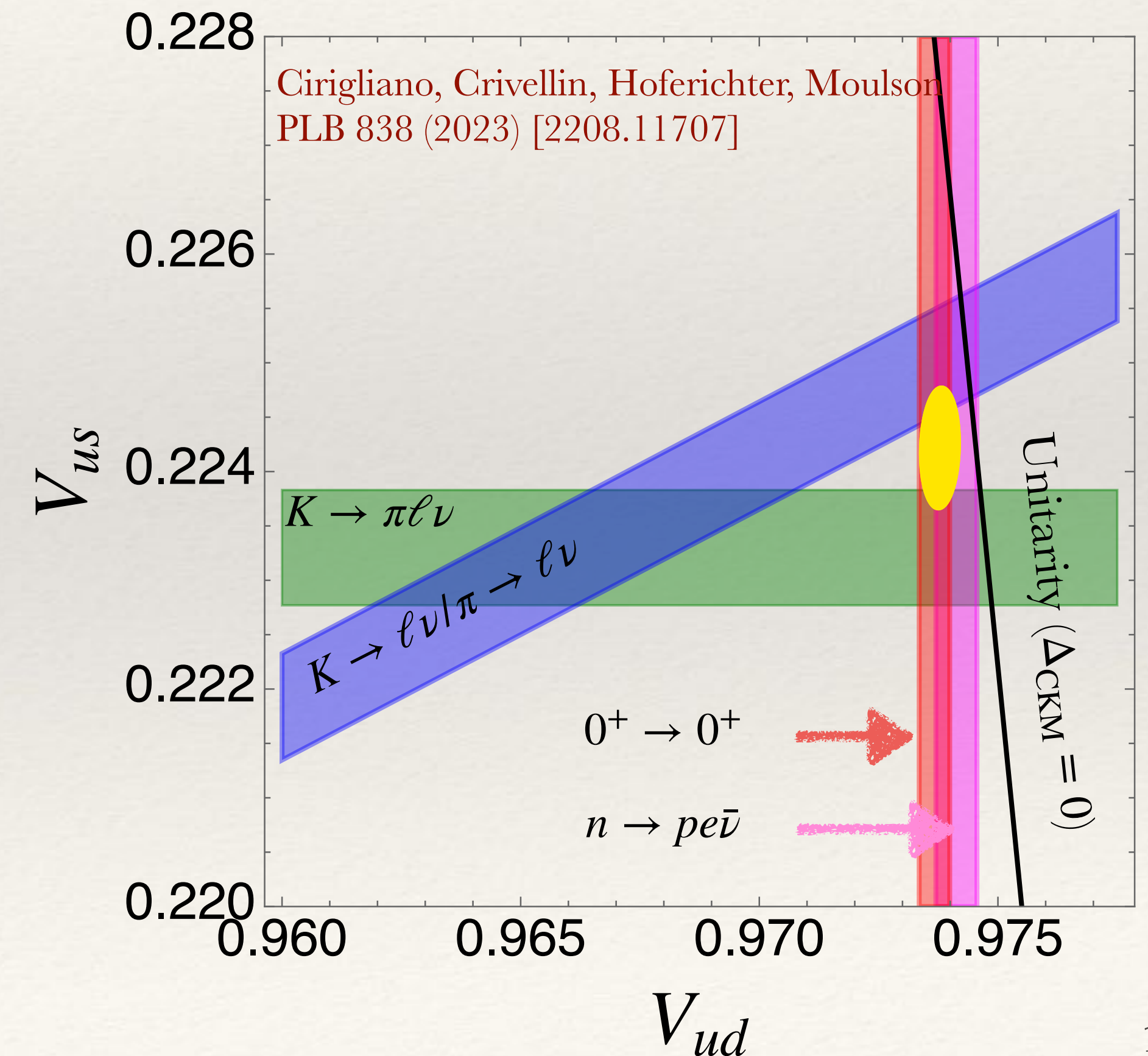
MDWF are more expensive (5 dimensional) — but really?

- F_K/F_π as a precision test of a new four flavor Domain Wall Fermion Action [2605.06560]
 - New four-flavor (u,d,s,c) MDWF action (2+1+1) generated
 - Grid was used to make configurations (USQCD HEP SciDAC)
 - QUDA used to perform “measurement” calculations
 - Perlmutter used to make 30 ensembles (15 months) — all of them now freely available via Globus (NERSC ERCAP - thank you Keith)
- F_K/F_π 2nd most precise LQCD result — very scrutinizing



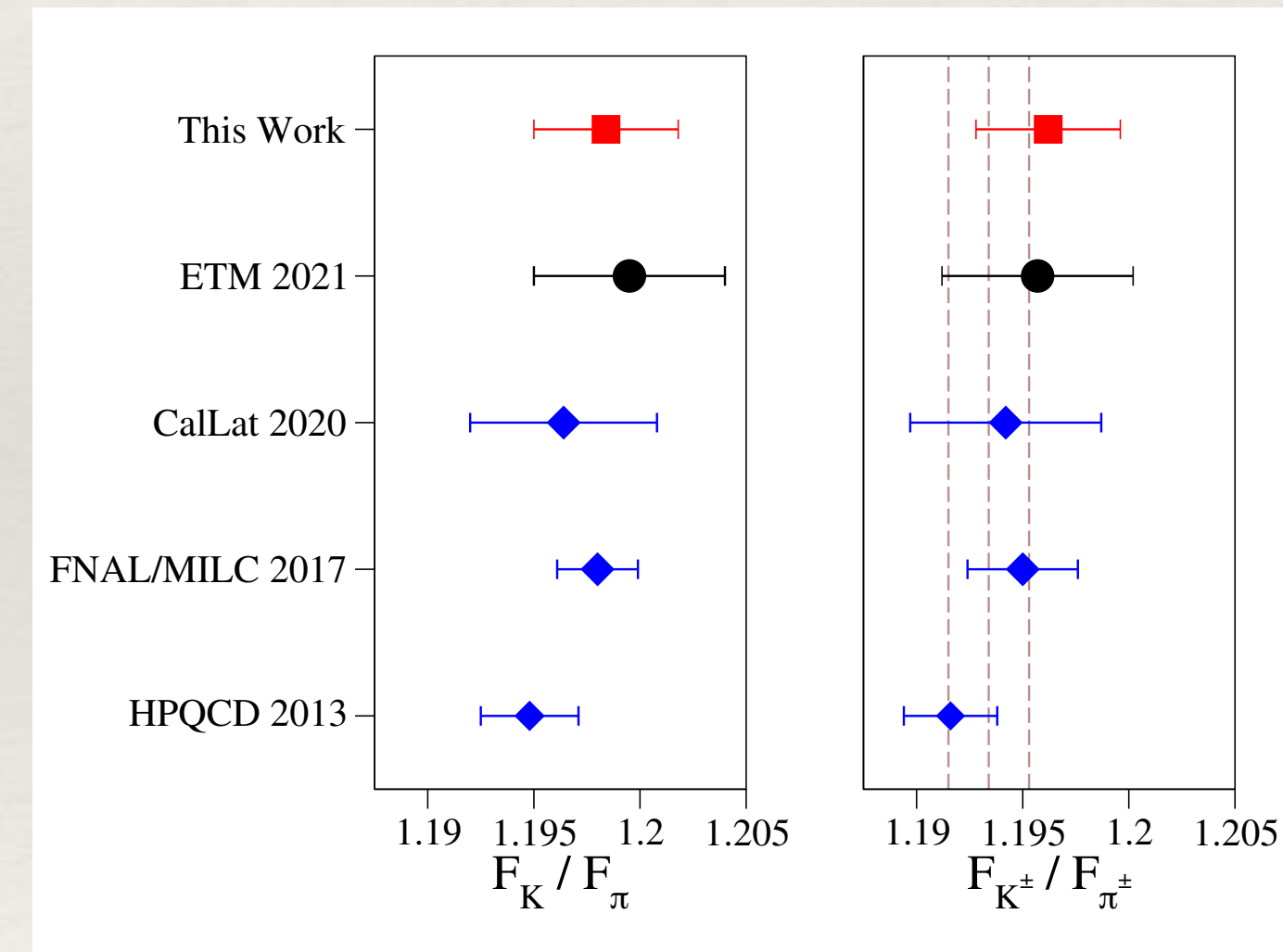
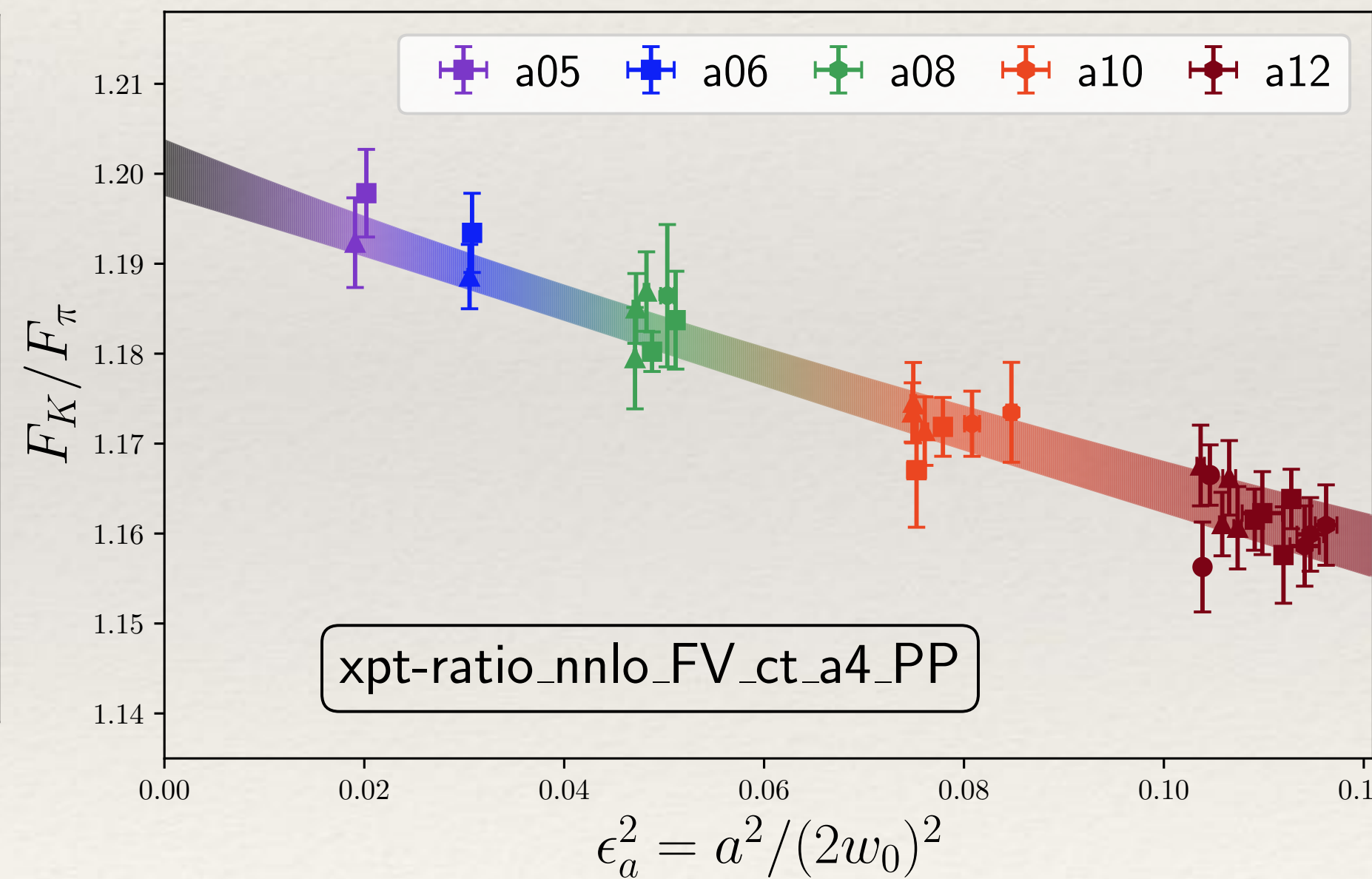
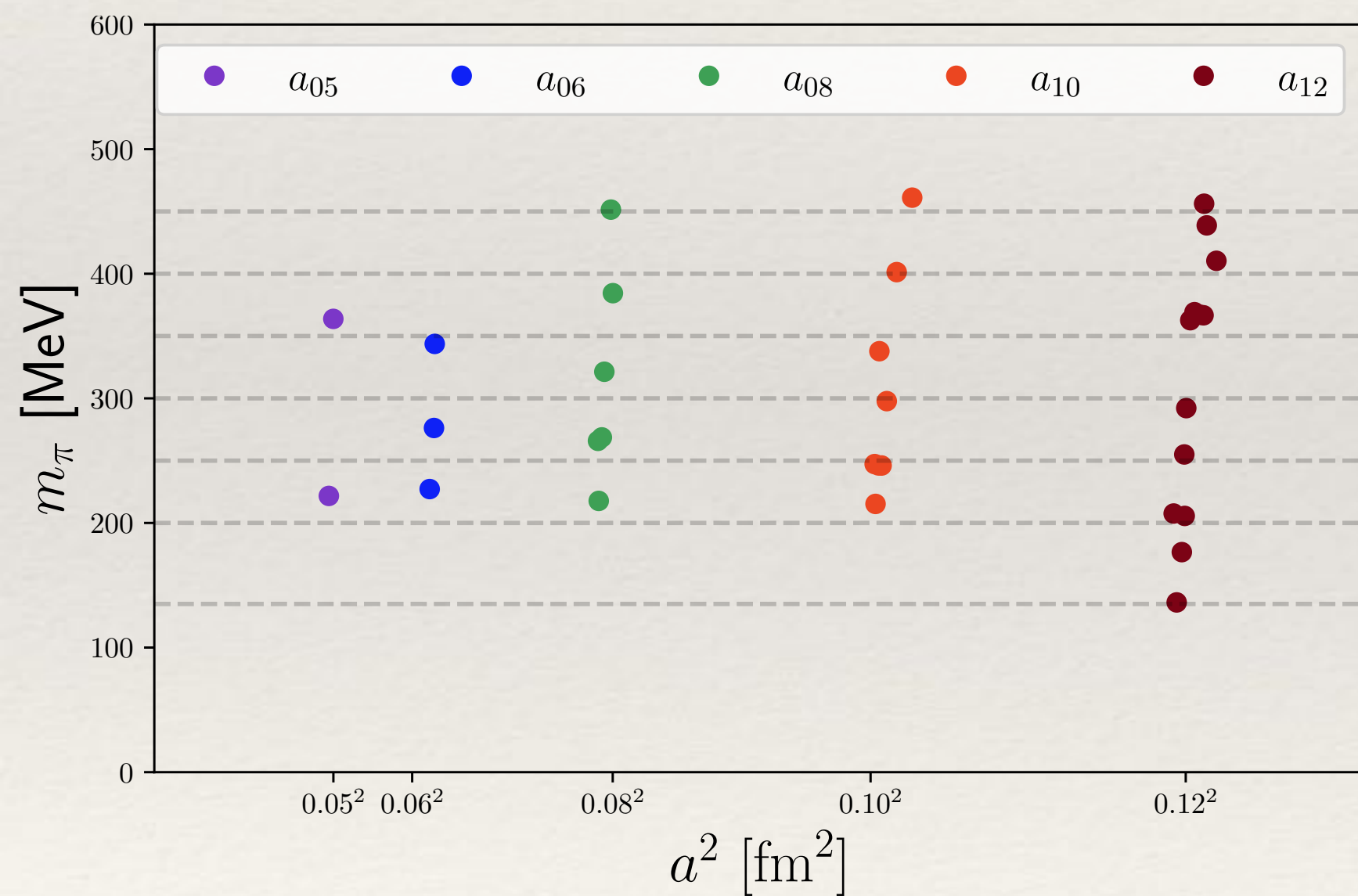
The diagram shows two Feynman diagrams for meson decays. The top diagram is for a K meson (red oval) decaying into a lepton $\ell_{\mu,e}$ and an antineutrino $\bar{\nu}_{\mu,e}$. The bottom diagram is for a π meson (red oval) decaying into a lepton $\ell_{\mu,e}$ and an antineutrino $\bar{\nu}_{\mu,e}$. Both diagrams feature a yellow star representing a four-flavor Domain Wall Fermion action vertex. The diagrams are separated by a horizontal line, and the entire expression is enclosed in large square brackets with a superscript 2.

$$= \frac{|V_{us}|^2}{|V_{ud}|^2} \frac{F_K^2}{F_\pi^2} f_{\text{kin.}} (1 + \delta_{\text{iso.}})$$



MDWF are more expensive (5 dimensional) — but really?

- F_K/F_π as a precision test of a new four flavor Domain Wall Fermion Action [2605.06560]
- New four-flavor (u,d,s,c) MDWF action (2+1+1) generated
- Grid was used to make configurations (USQCD HEP SciDAC)
- QUDA used to perform “measurement” calculations
- Perlmutter used to make 30 ensembles (15 months) — all of them now freely available via Globus (NERSC ERCAP - thank you Keith)
- F_K/F_π 2nd most precise LQCD result — very scrutinizing



- Spectroscopy tests underway — then onto many new projects: $K \rightarrow \pi \ell \nu$, $N \rightarrow N\pi, \Delta, \text{NN}, \dots$

Conclusions & Outlook

❑ Personal benefits:

- ❑ SciDAC has been critical for my research since SciDAC-I (eg. Chroma software suite)
- ❑ SciDAC-III (CalLat) cemented a new lattice QCD effort/group at LBNL

❑ Scientific benefits:

- ❑ Sustained development of optimized community libraries enabling efficient utilization of supercomputers — QUDA, Chroma, Grid, ...
- ❑ Lattice QCD has started making important predictions of single-nucleon properties (ie fully extrapolated to physical pion mass, continuum limit)
 $g_A, F_A(Q^2), f_i(x), \dots$
- ❑ We are on the cusp or making NN predictions at the physical point, as well as important inelastic nucleon structure
 $NN(m_\pi \lesssim 280 \text{ MeV}), \langle N\pi, \Delta | j_\mu^A | N \rangle, \dots$
- ❑ We've made significant progress exploring the use of “chiral fermions” (MDWF) for applications to nuclear physics problems — chiral symmetry is central to our understanding of low-energy QCD

Thank You!